

A Plücker-Derived Mass Operator for Exactly Unitary Dirac Quantum Walks

Phase Transport, Free Spectra, and a Verified Regulator Boundary

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Abstract

Standard Dirac quantum walks take their mass coin as an independent parameter. Let $\psi, \phi \in \mathbb{C}^2$ represent two future-null rank-one momenta, set $P_{\text{tot}} = \psi\psi^\dagger + \phi\phi^\dagger$, and let $z = \psi \wedge \phi$. Cauchy–Binet gives

$$\det P_{\text{tot}} = |z|^2.$$

Relative to a fixed chiral grading Γ , the oriented area determines the normalized odd Hermitian mass operator

$$B_z = \begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}, \quad B_z^2 = \det(P_{\text{tot}})\mathbf{1}, \quad (k\Gamma + B_z)^2 = (k^2 + \det P_{\text{tot}})\mathbf{1}.$$

We prove that this assignment is the unique positively homogeneous, phase-equivariant map with the displayed grading and normalization. The total momentum determines the orbit radius $|z| = \sqrt{\det P_{\text{tot}}}$; an oriented null factorization selects a representative B_z of that orbit. The positive rest energy is $|z|$, the spectral band gap at rest is $2|z|$, and both vanish exactly when the spinors are collinear. The construction derives the operator and its zero locus, but does not select an absolute mass scale.

Exponentiating B_z gives an exactly unitary null-step walk. Opposite turns carry z and $\bar{z}$, fixed-momentum symbols converge to Dirac flow uniformly on bounded momentum and mass sets at rate $O(1/n)$, and local chiral changes of basis induce exact endpoint links. A common phase is removable, but transported relative phase is not. For two equal-modulus sites coupled by an edge phase w , the exact free Hamiltonian has squared energies

$$E_\pm^2 = |z_L|^2 + t^2 \pm t |z_R - (\bar{w})^2 z_L|.$$

The mismatch modulus is invariant under independent local chiral-frame changes, and the lower level vanishes exactly at $t = |z_L|$ and $z_R = -(\bar{w})^2 z_L$. Thus null-spinor orientation supplies a free one-particle observable after transport. An independently assigned complex mass field can reproduce the same local phase physics; the Plücker contribution is its finite equivariant derivation, its determinant relation, and its collinearity zero locus from one primitive spinor input.

The finite statements and their negative controls are formalized in Lean 4. We also prove that the tested successive-axis $3 + 1$ extension retains a classified family of zero- and π -quasienergy aliases that no momentum-independent onsite coin removes. An alias-free exactly local $3 + 1$ regulator, a complete interacting field dynamics, and absolute scale selection remain open.

How to read the status marks. Origin and verification are independent. **CLASSICAL** identifies established mathematics formalized here; **NEW** identifies a new theorem or construction; **REALIZATION** identifies an explicit model. **KERNEL** means the stated finite claim is checked by the pinned

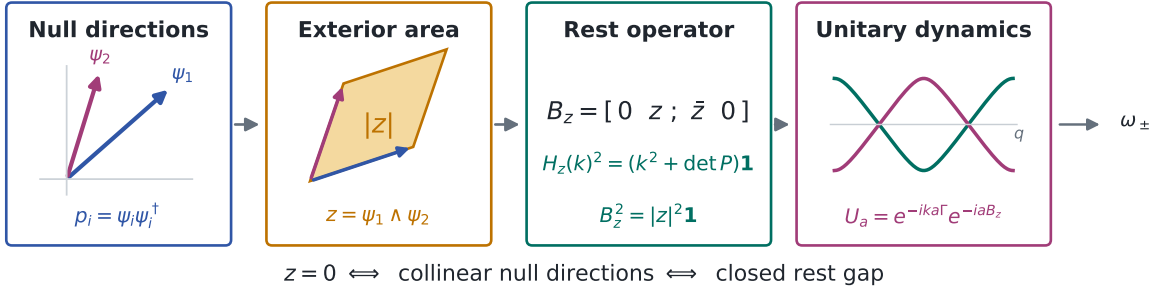


Figure 1: The complete finite construction. The input is a pair of null spinors, not a preassigned mass angle. The complex area fixes both the rest spectrum and the orientation phase of the unitary turn; collinearity is the exact gap-closing boundary.

Lean kernel; **PROOF** means a complete mathematical proof is given here but its dedicated Lean declaration is pending; **OPEN** marks a named unsolved problem. Physical interpretations are never promoted by a verification mark.

1 Introduction: one invariant, one operator, one walk

The sum of lightlike momenta need not be lightlike. In the spinor representation, each future null momentum is a rank-one positive matrix $\psi\psi^\dagger$. A collection of such momenta remains on the null boundary exactly while its spinors occupy one complex direction. Once two directions open, their exterior area is nonzero and the sum moves into the timelike interior.

This familiar geometry contains a sharper operator statement. The total momentum determines the radius $|z| = \sqrt{\det P_{\text{tot}}}$ of an odd-Hermitian mass orbit. An oriented null factorization retains the phase of $z = \psi \wedge \phi$ and therefore selects a representative B_z of that orbit. After a chiral grading and one normalization are fixed, equivariance makes the assignment unique. The operator squares to the invariant momentum determinant, anticommutes with the velocity grading, and exponentiates to an exact unitary turn gate. Transport between local chiral frames then converts relative orientation into a gauge-invariant spectral mismatch.

The null spinors in this statement are factors of a momentum representation, not additional asymptotic particles. The construction does not predict an absolute mass scale; a common mass phase is a basis choice; and covariance of the input map under $\text{SL}(2, \mathbb{C})$ does not make the finite lattice dynamics exactly Lorentz invariant. These boundaries are part of the claim, not qualifications added after it.

Synthesis theorem. Let $\psi, \phi \in \mathbb{C}^2$, set $P_{\text{tot}} = \psi\psi^\dagger + \phi\phi^\dagger$, $z = \psi \wedge \phi$, and $\mu = |z|$. Then

$$\det P_{\text{tot}} = \mu^2, \quad B_z = \begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}, \quad B_z^2 = \mu^2 \mathbf{1}.$$

Writing $\Gamma = \text{diag}(1, -1)$ and $H_z(k) = k\Gamma + B_z$, one has

$$\Gamma B_z + B_z \Gamma = 0, \quad \boxed{H_z(k)^2 = (k^2 + \det P_{\text{tot}}) \mathbf{1}}.$$

Thus the positive rest energy is $\mu = \sqrt{\det P_{\text{tot}}}$, and the gap closes exactly on the collinear null boundary. Multiplying z by a unit phase changes B_z by unitary conjugation, so P_{tot} determines the spectral orbit rather than an oriented representative. The factorization retains the phase needed by directed amplitudes and transported comparisons. NEW [KERNEL](#)

The precise novelty claim. Complex Dirac masses, mass textures, and gauge-covariant quantum walks are not new individually [?, ?]. The contribution here is the finite equivariant map from an oriented null-spinor factorization to the complex mass operator, together with its exact determinant relation, unitary-history dictionary, transported free spectrum, convergence bound, and formal verification. An assigned complex mass field can reproduce the same one-particle phase phenomena; the new content is that modulus, phase, and the collinearity zero locus arise coherently from the same null-spinor input. The construction neither predicts the absolute scale nor removes the tested $3 + 1$ aliases.

Three claims organize the paper.

1. **Geometry and naturality.** P_{tot} determines the radius of the odd-Hermitian mass orbit; an oriented factorization selects its representative. Positive homogeneity, phase covariance, the chiral grading, and one normalization uniquely select $z \mapsto B_z$.
2. **Dynamics and observability.** B_z generates an exactly unitary walk, directed histories, and a uniform $O(1/n)$ continuum-symbol bound. Local phase transport produces the gauge-invariant mismatch $|z_R - (\bar{w})^2 z_L|$, which splits an exact free spectrum.
3. **Regulator boundary.** The tested successive-axis $3 + 1$ extension has completely classified zero- and π -quasienergy aliases; no momentum-independent onsite coin removes them. This is a theorem about that architecture, not a universal no-go for local $3 + 1$ walks.

The main logical chain is therefore

$$\text{null geometry} \longrightarrow B_z \longrightarrow \text{unitary histories} \longrightarrow \text{transport} \longrightarrow \text{relative-phase spectrum}.$$

The extended regulator and Fock calculations are retained as boundary and companion results, but they do not carry the central novelty claim.

The result sits between spinor-helicity kinematics and discrete relativistic quantum dynamics [?, ?]. Cauchy–Binet and the Pauli map supply the classical foundation. Feynman’s checkerboard and modern quantum walks show how a massive Dirac propagator can arise from null steps and direction changes [?, ?, ?, ?, ?]. Our contribution is the exact finite map from the oriented spinor area to the odd rest operator and then to a controlled unitary walk, together with a machine-checked dependency boundary.

Construction	Exact local unitary	Exact finite histories	Mass input	Dimension	Continuum control	Machine checked
Feynman checkerboard [?, ?]	not its object	yes	assigned corner weight	1 + 1	propagator limit	no
Dirac QCA [?, ?]	yes	yes	automaton parameter	1 + 1	channel/wave-packet bounds	no
Split-step DCA [?]	yes	not central	tunable coin angle	1 + 1	Dirac limit	no
Weyl/Dirac QCA [?]	yes	model dependent	Weyl coupling	up to 3 + 1	relativistic limit	no
Tetrahedral Dirac walk [?]	yes	not central	assigned $e^{-im\gamma^0}$ coin	3 + 1 tetrahedra	first-order Dirac limit	no
Successive-axis walk [?]	yes	not central	assigned mass parameter	3 + 1	first-order relativistic limit	no
This work	yes	yes	$z = \psi \wedge \phi$	1 + 1; finite 3 + 1 lift	uniform $O(1/n)$ symbols; three-torus L^2	yes

Table 1: Structural comparison, not a priority ranking. The closest QCA works have stronger field-theoretic or higher-dimensional scope. The present distinction is the geometrically generated complex rest operator, the exact kernel dictionary, and their formal verification.

Nearest constructions and the exact novelty claim

For constant z , the update in this paper is unitarily equivalent to the standard 1 + 1 split-step Dirac walk. We do not claim otherwise. What is new is the input map and its naturality: the coin is generated by the complex Plücker coordinate of null momentum factors, its modulus gives the rest energy, its phase acts by internal conjugacy and survives in oriented turn amplitudes, and the resulting dependency chain is machine checked. Table ?? positions that claim against the closest constructions.

The Pavia programme moreover derives the Weyl and Dirac automata from informational first principles [?]; the derivation offered here is geometric rather than axiomatic, and the two are answers to different questions.

The kinematic layer itself has long roots, and we claim none of it. Weyl–van der Waerden and massive spinor-helicity formalisms routinely decompose a massive momentum into two null spinors, with $\det K = \lambda_1 \lambda_2 = k^2$ appearing explicitly in the eigen-decomposition of the momentum bispinor [?], and the bold-spinor formalism of Arkani-Hamed, Huang, and Huang [?] makes the two-null-spinor description of massive states the organizing principle of amplitude kinematics. Those formalisms use the decomposition as covariant bookkeeping for scattering amplitudes. The present paper starts where that bookkeeping stops: the wedge coordinate is promoted to the off-diagonal entry of a canonical odd Hermitian operator, the operator is classified (Theorem ??), exponentiated into an exactly unitary walk, tracked through directed histories and a local connection, and every finite statement in that chain is kernel-checked. The amplitude formalisms neither assert nor need such a dynamical chain; the overlap is the classical 2×2 Cauchy–Binet layer, which we label CLASSICAL throughout.

Exact consequences for published constructions

Beyond structural comparison, several of this paper’s kernel-checked obstructions bear directly on published proposals. We collect the exact statements here; each is proved in the section cited, and none is a new claim.

- **Stay-put walks (Gupta–Short).** The doubler- and pseudo-doubler-free family of Gupta and Short [?] is exactly of the degree-one Laurent form covered by the stationary-amplitude no-go (?), so its tangent generator cannot be an involution: the stay-put freedom that removes the doublers is purchased exactly by relinquishing the involutory unit-speed Dirac tangent (corollary in the text around Eq. (??); their Eqs. (29)–(30) and (37)). Both programs share the residual open problem their Appendix F records: extraneous Weyl-like low-energy solutions at isolated momenta.
- **Onsite repairs of successive-axis architectures.** For the four-channel, range-one, single-factor-per-axis 3 + 1 architecture — the class containing the successive-axis composition

formalized here, following [?] — the three even-parity origin aliases are exact, and the all-coins alias theorem (??) proves that *no* momentum-independent onsite coin, the complex Plücker mass coin included, removes any of them. Extra range, coupled substeps, or enlarged cells are structurally required, not optional. Exact two-step temporal blocking closes only the mass subgroup, so composition-order effects in such architectures are exact, not perturbative artifacts.

- **Scope relative to action-based no-gos.** The Nielsen–Ninomiya theorem [?] governs chirally-invariant local *action/Hamiltonian* discretizations, and the minimally-doubled Karsten–Wilczek/Boriçi–Creutz families live in that class. The obstructions here are the strictly-unitary *discrete-time* analogues (exact one-step unitarity, origin normalization, involutory tangent); neither family of results implies the other, and the two boundaries constrain a $3 + 1$ regulator from logically independent directions.

A complementary July 2026 analysis of the $1 + 1$ chiral Dirac QCA treats doubling through Green functions and a modified discretization [?]. Its scope differs from the exact discrete-time $3 + 1$ alias classification here, but it sharpens the current comparison set for any successor regulator.

The reparametrization test

At constant nonzero z , the one-particle spectrum alone cannot distinguish this construction from a Dirac walk whose mass was assigned as $m = |z|$. Moreover, an independently assigned complex mass field can reproduce the same one-particle phase and transport phenomena. The claim of added content must therefore survive both baselines. Table ?? states the test explicitly.

Model	What it can reproduce	What remains external or newly derived
Assigned real $m(x)$	Constant dispersion after $m = z $; one real turn weight	No local phase, endpoint link, or transported-phase splitting
Assigned complex $M(x)$	The same local phase, directed $M/\bar{M}$ histories, endpoint links, and free spectral split	The complex field, its zero locus, and its relation to momentum are supplied independently
Derived $z(x) = \psi(x) \wedge \phi(x)$	Every complex point value, with the same one-particle phase phenomena as $M(x)$	The complex field now inherits $ z ^2 = \det P_{\text{tot}}$, the exact collinearity zero locus, and its orientation from one spinor factorization

Table 2: What is and is not gained by deriving the complex mass field. An assigned complex mass is the correct one-particle baseline: the new content is the field’s geometric provenance, determinant relation, and exact collinearity zero locus, not the existence of phase physics by itself. None of the three models predicts the absolute mass scale, and all retain regulator-dependent ultraviolet structure.

This comparison also fixes the paper’s ceiling. Its centerpiece is a one-particle $1 + 1$ walk with a new geometric input and a rigorous finite chain; Section ?? adds a controlled finite $3 + 1$ local extension. It is not yet a new QCA classification, an interacting QCA, or a field theory; its exact high-symmetry audit in fact exposes corner aliases and a mass-independent body-center no-gap obstruction that a successor regulator must remove.

2 Null-spinor area and the future cone

2.1 From rank-one momenta to area

Let $\psi_1, \dots, \psi_N \in \mathbb{C}^2$ and define

$$M = [\psi_1 \ \cdots \ \psi_N], \quad P = MM^\dagger = \sum_{i=1}^N \psi_i \psi_i^\dagger. \quad (1)$$

Each summand is positive semidefinite and has rank at most one. Under the mostly-minus Pauli map, it is a future null four-momentum. The determinant of P is the Lorentzian square of the total momentum.

Theorem 2.1 (Null-spinor area identity). *For the matrix in (??),*

$$\det P = \left\| \bigwedge^2 M \right\|_{\text{HS}}^2 = \sum_{i < j} |\det [\psi_i \ \psi_j]|^2. \quad (2)$$

Consequently $\det P \geq 0$, and $\det P = 0$ if and only if the occupied columns span a subspace of dimension at most one.

CLASSICAL KERNEL

Proof. Cauchy–Binet applied to $MM^\dagger$ expresses its determinant as the sum of the squared moduli of all 2×2 minors of M . Those minors are exactly the coordinates of $\bigwedge^2 M$. Every summand is nonnegative, so the sum vanishes exactly when every pairwise wedge vanishes, equivalently when $\text{rank } M \leq 1$. $\square$

Define

$$\mu := \left\| \bigwedge^2 M \right\|_{\text{HS}} = \sqrt{\det P}. \quad (3)$$

The quantity μ is the first-order rest energy; μ^2 is the Lorentzian quadratic invariant. The exterior norm makes the geometric word “area” literal rather than metaphorical.

For readers who prefer spatial directions, write the null momenta as $p_i = E_i(1, \mathbf{n}_i)$ with $E_i \geq 0$ and $|\mathbf{n}_i| = 1$. With the same mostly-minus normalization,

$$\mu^2 = 2 \sum_{i < j} E_i E_j (1 - \mathbf{n}_i \cdot \mathbf{n}_j). \quad (4)$$

Mass squared is therefore a positive sum of directional disagreements. It is zero only when every occupied direction agrees; no cancellation between different pairs is available.

2.2 Why the determinant is canonical

The determinant is not selected merely because it gives the desired answer. On the physical space of Hermitian 2×2 matrices it is the unique real quadratic form, up to scale, that vanishes on the full rank-one positive boundary.

Proposition 2.2 (Null-boundary uniqueness). *Let q be a real quadratic form on $\text{Herm}(2)$. If $q(\psi\psi^\dagger) = 0$ for every $\psi \in \mathbb{C}^2$, then $q(P) = c \det P$ for some $c \in \mathbb{R}$.*

CLASSICAL KERNEL

Proof. Identify $P \in \text{Herm}(2)$ with $(t, \mathbf{x}) \in \mathbb{R}^{1,3}$ so that $\det P = t^2 - |\mathbf{x}|^2$. Write $q(t, \mathbf{x}) = at^2 + 2t \mathbf{b} \cdot \mathbf{x} + \mathbf{x}^T C \mathbf{x}$, with C symmetric. On the future null cone, homogeneity reduces the hypothesis to $q(1, \mathbf{n}) = 0$ for every unit vector $\mathbf{n}$. Comparing $\mathbf{n}$ with $-\mathbf{n}$ gives $\mathbf{b} = 0$. Hence $\mathbf{n}^T C \mathbf{n} = -a$ on the unit sphere, which forces $C = -aI$ by polarization. Thus $q = a(t^2 - |\mathbf{x}|^2) = a \det P$. $\square$

The Lean theorem uses the four real coordinates of a Hermitian matrix and nine explicit rank-one spinor probes to eliminate every quadratic coefficient except one multiple of $pq - r^2 - s^2$. The older real-symmetric slice and a form that fails the full-boundary hypothesis remain as independent controls.

Corollary 2.3 (First-order uniqueness). *Suppose g is nonnegative and homogeneous of degree one on the future cone, and $g(P)^2$ is a quadratic form vanishing on the rank-one boundary. Then $g(P) = c\sqrt{\det P}$ for some $c \geq 0$.*

CLASSICAL PROOF

This is the logical bridge from a canonical mass-squared invariant to a canonical positive rest energy. The normalization $c = 1$ fixes units.

3 The Plücker mass operator

Two copies of $\mathbb{C}^2$ play different roles. Let $S \simeq \mathbb{C}^2$ be the null-spinor space and let $C_{\text{ch}} = C_+ \oplus C_-$ be the two-channel chiral space of the walk. The area lies in the one-dimensional line $\bigwedge^2 S$, whereas the rest operator acts on C_{ch} . Choosing an oriented volume form on S , the grading $C_+ \oplus C_-$, and a unit normalization identifies $\bigwedge^2 S$ with the weight-one off-diagonal line $\text{Hom}(C_-, C_+)$. All uses of “canonical” below are relative to exactly these choices.

For two spinors, retain the complex orientation data rather than immediately discarding it into an absolute square:

$$z := \psi \wedge \phi = \psi_0 \phi_1 - \psi_1 \phi_0, \quad P_{\text{tot}} := \psi \psi^\dagger + \phi \phi^\dagger. \quad (5)$$

Theorem ?? gives $|z|^2 = \det P_{\text{tot}}$. Define

$$B_z := \begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}, \quad \Gamma := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (6)$$

Equivalently,

$$B_z = (\text{Re } z)\sigma_x - (\text{Im } z)\sigma_y. \quad (7)$$

The map does not restrict the pointwise complex mass parameter: with the standard basis e_1, e_2 of S , $\psi = e_1$ and $\phi = ze_2$ give $\psi \wedge \phi = z$ for every $z \in \mathbb{C}$. Its content is geometric provenance, the determinant relation, and the exact zero locus, not a smaller set of allowed complex masses.

Theorem 3.1 (Plücker-derived rest operator). *The matrix B_z is Hermitian and odd for the grading Γ . More precisely,*

$$B_z^\dagger = B_z, \quad \Gamma B_z + B_z \Gamma = 0, \quad B_z^2 = |z|^2 \mathbf{1} = \det(P_{\text{tot}}) \mathbf{1}. \quad (8)$$

For

$$H_z(k) := k\Gamma + B_z, \quad (9)$$

one has

$$\boxed{H_z(k)^2 = (k^2 + \det P_{\text{tot}}) \mathbf{1}}. \quad (10)$$

The rest eigenvalues are $\pm|z|$, and $B_z = 0$ exactly when the two nonzero spinor directions are collinear.

NEW [KERNEL](#)

Proof. Hermiticity is immediate from the two off-diagonal entries. Direct multiplication gives $B_z^2 = \text{diag}(z\bar{z}, \bar{z}z) = |z|^2 \mathbf{1}$, while the diagonal signs in Γ give the anticommutator zero. Expanding $H_z(k)^2$ leaves $k^2 \mathbf{1} + B_z^2$ because the mixed term vanishes. Finally $B_z = 0$ if and only if $z = 0$, and a two-spinor wedge vanishes exactly on projective collinearity. Every step of this proof is represented by a general Lean declaration, not by evaluation of a fixed example. $\square$

Theorem ?? is stronger than assigning the scalar $\det P_{\text{tot}}$ to an otherwise independent Dirac mass slot. The odd operator is constructed from the oriented Plücker coordinate itself; its square derives the quadratic invariant. The distinction between powers is now fixed:

Layer	Natural quantity
Null-spinor kinematics	$\mu^2 = \det P_{\text{tot}} = z ^2$
Odd mass operator	B_z , with $B_z^2 = \mu^2 \mathbf{1}$
Quadratic costs and Hessians	μ^2
Dirac rest energy	$\mu = z $
Finite unitary turn angle	$a\mu$
Exact corner/stay ratio	$\tan(a\mu)$

3.1 Phase covariance, naturality, and the mass orbit

If $u \in \mathbb{C}$ has $|u| = 1$, let $W_u = \text{diag}(u, 1)$. Then

$$W_u B_z W_u^\dagger = B_{uz}, \quad W_u W_u^\dagger = \mathbf{1}. \quad (11)$$

Both identities are kernel-checked. Thus the phase of z rotates the internal basis but cannot change the rest spectrum.

Section ?? uses the determinant-one chiral frame $\widehat{W}_b = \text{diag}(b, \bar{b})$ instead. It sends $z \mapsto b^2 z$. For $u = b^2$, $\widehat{W}_b$ differs from W_u only by a central phase, so the two conventions induce the same conjugation of B_z .

The input map is also covariant under the spacetime spin-group action. If $A \in \text{SL}(2, \mathbb{C})$ acts on both spinors by left multiplication, then

$$(A\psi) \wedge (A\phi) = \det(A)(\psi \wedge \phi) = z, \quad B_{(A\psi) \wedge (A\phi)} = B_z. \quad (12)$$

This is an algebraic covariance statement for the input map under the spin cover of the proper-orthochronous Lorentz group; it is [KERNEL](#). It does not assert exact Lorentz symmetry of the finite lattice update. General $A \in \text{GL}(2, \mathbb{C})$ acts with the expected determinant weight; that weighted extension is an algebraic [PROOF](#) corollary, not a separate Lean declaration in the present development.

Theorem 3.2 (Classification of odd Hermitian rest operators). *Let $B \in M_2(\mathbb{C})$ be Hermitian and satisfy $\Gamma B + B\Gamma = 0$. Then there is a unique $w \in \mathbb{C}$ such that*

$$B = B_w = \begin{pmatrix} 0 & w \\ \bar{w} & 0 \end{pmatrix}, \quad B^2 = |w|^2 \mathbf{1}. \quad (13)$$

Consequently the odd Hermitian square roots of $\mu^2 \mathbf{1}$ form one phase circle $w = \mu u$, $|u| = 1$, and for $\mu > 0$ they are all related by the conjugacies in (??). Specifying the oriented Plücker coordinate z selects one member of that orbit.

NEW [KERNEL](#) for the normal form and square; [PROOF](#) for the phase-orbit corollary.

Proof. Oddness forces both diagonal entries of B to vanish, and Hermiticity forces the lower-left entry to be the complex conjugate of the upper-right entry $w = B_{01}$. The square is then $|w|^2 \mathbf{1}$. If it equals $\mu^2 \mathbf{1}$, then $|w| = \mu$; for $\mu > 0$, write $w = \mu u$ with $|u| = 1$ and apply [\(??\)](#). $\square$

The classification theorem describes the target space. The following naturality statement is what makes the assignment itself unique.

Theorem 3.3 (Equivariant uniqueness of the normalized rest map). *Fix $\Gamma = \text{diag}(1, -1)$ and $W_u = \text{diag}(u, 1)$. Suppose $F : \mathbb{C} \rightarrow M_2(\mathbb{C})$ satisfies:*

- (i) $F(z)$ is Hermitian and $\Gamma F(z) + F(z) \Gamma = 0$;
- (ii) $F(0) = 0$ and $F(rz) = rF(z)$ for every $r \geq 0$;
- (iii) $F(uz) = W_u F(z) W_u^\dagger$ whenever $|u| = 1$;
- (iv) $F(1) = \sigma_x$.

Then $F(z) = B_z$ for every $z \in \mathbb{C}$.

[PROOF](#) The theorem records the precise covariance class and normalization in which the words *natural* and *uniquely selected* are used; a dedicated Lean declaration is not claimed.

Proof. The case $z = 0$ is an assumption. Otherwise write $z = ru$ with $r = |z| > 0$ and $|u| = 1$. Positive homogeneity, equivariance, and normalization give

$$F(z) = rF(u) = rW_u F(1) W_u^\dagger = rW_u \sigma_x W_u^\dagger = B_{ru} = B_z.$$

$\square$

This also controls the nonuniqueness of the null decomposition. If $M = [\psi \ \phi]$ has full rank and $M' = MR$ with $R \in U(2)$, then $M' M'^\dagger = M M^\dagger$ and $\det M' = (\det R) \det M$. Since $|\det R| = 1$, the two mass operators are related by [\(??\)](#). The unitary-equivalence class of B_z depends only on P_{tot} ; an oriented factorization retains the phase needed by individual turn amplitudes. [CLASSICAL KERNEL](#)

The distinction matters physically. The spectrum is decomposition independent, while an amplitude can remain sensitive to orientation. The construction preserves exactly the information needed by a path sum and forgets exactly the information a rest energy cannot observe.

4 Gauge-covariant phase transport and a free spectral observable

At one point, every B_z with the same $|z|$ belongs to the same chiral conjugacy orbit. Relative orientation becomes observable only after local representatives are compared through transport. Consider two chiral spaces $C_L \oplus C_R \simeq \mathbb{C}^4$. For $|w| = 1$ and $t \geq 0$, define

$$D(w) = \text{diag}(w, \bar{w}), \quad T(t, w) = t \Gamma D(w), \quad H_2(z_L, z_R, w, t) = \begin{pmatrix} B_{z_L} & T(t, w) \\ T(t, w)^\dagger & B_{z_R} \end{pmatrix}. \quad (14)$$

In the ordered basis (L_+, L_-, R_+, R_-) this is

$$H_2 = \begin{pmatrix} 0 & z_L & tw & 0 \\ \bar{z}_L & 0 & 0 & -t\bar{w} \\ t\bar{w} & 0 & 0 & z_R \\ 0 & -tw & \bar{z}_R & 0 \end{pmatrix}. \quad (15)$$

Independent local chiral-frame changes $b_L, b_R \in U(1)$ act by

$$z_L \mapsto b_L^2 z_L, \quad z_R \mapsto b_R^2 z_R, \quad w \mapsto b_L w \bar{b}_R. \quad (16)$$

Direct conjugation by $V_{b_L, b_R} = \text{diag}(b_L, \bar{b}_L, b_R, \bar{b}_R)$ gives the transformed Hamiltonian. The transported mismatch

$$\Delta := z_R - (\bar{w})^2 z_L \quad (17)$$

therefore transforms as $\Delta \mapsto b_R^2 \Delta$, so $|\Delta|$ is gauge invariant. The square on w reflects the half-phase carried by each chiral component.

Theorem 4.1 (Transported-phase spectral polynomial). *Assume $t \geq 0$, $|z_L| = |z_R| = \mu$, and $|w| = 1$. Then H_2 is Hermitian and*

$$\boxed{(H_2^2 - (\mu^2 + t^2)\mathbf{1})^2 = t^2 |\Delta|^2 \mathbf{1}.} \quad (18)$$

Consequently the squared energies are

$$E_{\pm}^2 = \mu^2 + t^2 \pm t|\Delta|. \quad (19)$$

When $t|\Delta| > 0$, each level has total multiplicity two. The lower level vanishes exactly when

$$t = \mu, \quad z_R = -(\bar{w})^2 z_L. \quad (20)$$

Formal verification. Hermiticity, the polynomial identity, the eigenvalue polynomial, trace balance, and the zero-mode equivalence are kernel checked in `PlueckerPhaseDefectSpectrum`; the displayed multiplicity formulation is the corresponding finite-dimensional spectral consequence.

The common-phase case is a necessary control. For $|b| = 1$,

$$V_b H_2(z_L, z_R, w, t) V_b^\dagger = H_2(b^2 z_L, b^2 z_R, w, t), \quad V_b = \text{diag}(b, \bar{b}, b, \bar{b}), \quad (21)$$

so a common phase cannot change the spectrum. The equal-modulus hypothesis in Theorem ?? is also load-bearing: the exact fixture $z_L = 1, z_R = 2, t = w = 1$ violates Eq. (??). The theorem does not hide amplitude variation inside a claimed phase effect.

This spectral effect is not unique to a Plücker parametrization. An assigned complex mass field $M(x)$ with the same local covariance law can reproduce it. An assigned real scalar mass cannot. What the construction adds is that the complex field, its modulus relation to P_{tot} , its collinearity zero locus, and its transport law arise from one oriented null-spinor input. This is the reparametrization test in its strongest baseline form.

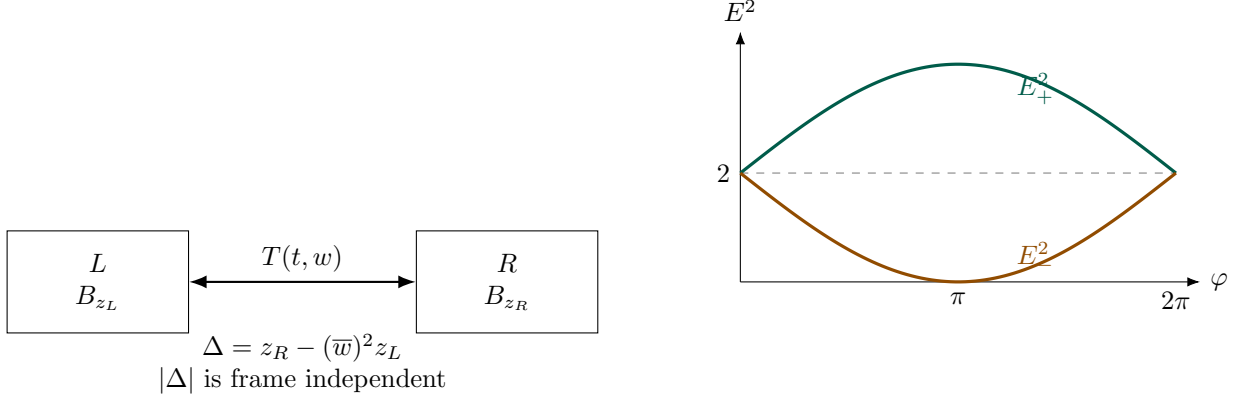


Figure 2: Left: local mass representatives are compared through an edge transport. Right: for $w = 1$, $z_L = 1$, $z_R = e^{i\varphi}$, and $t = 1$, $E_{\pm}^2 = 2 \pm 2|\sin(\varphi/2)|$. A common phase leaves both curves unchanged; transported relative phase splits the free spectrum and reaches a zero mode at $\varphi = \pi$.

4.1 Supplementary generalized closure and interaction seed

The two-spinor operator above is the visible corner of a single finite algebra. For *any* number of spinors $\psi_i = (u_i, v_i)$, the pairwise areas $z_{ij} = u_i v_j - u_j v_i$ assemble into the antisymmetric matrix $Z = uv^\top - vu^\top$, and the complete area budget is the Gram determinant, equal by an exact Lagrange identity to the total pairwise area, $\mu^2 = \sum_{i < j} |z_{ij}|^2$. The kernel-checked structure (`PlueckerRestOperatorGeneral`) is:

$$ZZ^\dagger Z = \mu^2 Z, \quad B_w = \begin{pmatrix} 0 & Z \\ Z^\dagger & 0 \end{pmatrix}, \quad B_w^3 = \mu^2 B_w, \quad (22)$$

with B_w odd Hermitian for the block grading, B_w^2 equal to μ^2 times a projector of trace exactly four for every n with $\mu^2 \neq 0$ — the rest gap $\pm\mu$ always lives on a four-dimensional support — the two-spinor case recovering $B_z^2 = \mu^2 \mathbf{1}$ on the whole space, and the massless boundary exact: $Z = 0 \iff \mu^2 = 0$. The spinor origin of the area data is load-bearing in a precise sense: an explicit non-decomposable antisymmetric control (two disjoint 2×2 blocks) satisfies every symmetry hypothesis and still obeys a proportional cubic closure, but with the wrong coefficient — $ZZ^\dagger Z = 1 \cdot Z$ while its total pairwise area is 2 (witness entry 1 against 2) — so what the spinor origin buys is the *coefficient identification*: the closure constant equals the area budget exactly for spinor-generated data. NEW `KERNEL`

The same closure governs the interaction layer. The supplied pair kick of the companion dynamics paper is, up to a global phase, the exact quarter-period pulse of a Hermitian even quartic CAR generator $K(z)$ obeying

$$K(z)^3 = |z|^2 K(z) \quad (23)$$

on its pair block (`PlueckerPairGenerator`, `KERNEL`), with the gate family closing into an explicit circle group. Equations (??) and (??) have the same finite statement shape — a cubic closure with the squared datum as coefficient — realized once by the odd one-particle rest operator of arbitrarily many null constituents and once by the even many-body coupling that reads the same complex datum. We state this as a coincidence of the shapes of two independently landed theorems, not as a unification: no lemma relates B_w to $K(z)$. The selection question — which coupling reads the phase — is resolved (the covariance-forced kick); what remains open is the distinct *dynamical* question, whether the free walk itself forces that interaction, which we list separately in the open problems.

What the coincidence does license is a sharper organizing principle for the program: every operator this construction has derived or supplied so far is annihilated by $X^3 - (\text{area datum})X$.

5 Exact unitary null histories

5.1 Exponentiating the derived rest term

For $\mu = |z| > 0$, set $\hat{B}_z = B_z/\mu$. Since $\hat{B}_z^2 = \mathbf{1}$,

$$e^{-iaB_z} = \cos(a\mu)\mathbf{1} - i\sin(a\mu)\hat{B}_z. \quad (24)$$

At $\mu = 0$ the expression is understood by continuity and equals $\mathbf{1}$. Combining the turn with the conditional null translation gives the momentum symbol

$$U_a(k; z) := e^{-iak\Gamma} e^{-iaB_z}. \quad (25)$$

Each factor is unitary. The two components translate in opposite null directions; only the odd factor changes direction.

The finite formula in (??) is itself closed under composition: writing it as $C_z(a)$, one has

$$C_z(a)C_z(a)^\dagger = \mathbf{1}, \quad C_z(a)C_z(b) = C_z(a+b). \quad (26)$$

The explicit rest eigenvectors, right-unitary decomposition theorem, and this one-parameter group law are all [KERNEL](#).

By phase covariance, $U_a(k; z)$ is unitarily conjugate to the real-mass walk with parameter μ . Its determinant is one and its trace is $2\cos(ka)\cos(\mu a)$. Therefore an eigenphase $e^{-i\omega a}$ obeys

$$\boxed{\cos(\omega a) = \cos(ka)\cos(\mu a)}. \quad (27)$$

At $k = 0$ the exact eigenphases are $e^{\mp ia\mu}$, so the positive principal-branch quasienergy is μ . Quasienergy is periodic modulo $2\pi/a$; no global unbranched equality is asserted. The real-mass spectral statement and oriented mass-coin group law are [KERNEL](#); the displayed spectral conjugacy is an immediate corollary of the kernel-checked phase covariance.

5.2 The exact corner ratio

Equation (??) gives two kinds of matrix element. A component that keeps its direction contributes $\cos(a\mu)$. A direction change contributes $-i\sin(a\mu)$ times z/μ or $\bar{z}/\mu$, according to its orientation. Hence an n -step history h with $c(h)$ turns has amplitude

$$A_a(h) = e^{i\Phi_k(h)} \cos(a\mu)^{n-c(h)} (-i\sin(a\mu))^{c(h)} \Xi_z(h), \quad (28)$$

where $\Phi_k(h)$ is the translation phase and $\Xi_z(h)$ is a product of unit-modulus orientation factors. Whenever $\cos(a\mu) \neq 0$,

$$A_a(h) = e^{i\Phi_k(h)} \cos(a\mu)^n (-i\tan(a\mu))^{c(h)} \Xi_z(h). \quad (29)$$

Proposition 5.1 (Unitary checkerboard correspondence). *After removal of the common factor $\cos(a\mu)^n$, the exactly unitary walk is a corner-weighted checkerboard sum with finite-spacing corner parameter $\tan(a\mu)$ and the orientation phase inherited from z . Under the opposite checkerboard sign convention, the parameter changes sign. In either convention,*

$$\tan(a\mu) = a\mu + O(a^3), \quad (30)$$

but $\tan(a\mu)$ and $a\mu$ are not identical at finite spacing.

CLASSICAL for the coin parametrization; NEW [KERNEL](#) for the exact two-weight recursion scaling and the separate transfer-matrix identification; NEW [KERNEL](#) for the directed complex orientation bookkeeping in the same path sum.

Proof. Insert the two matrix elements from (??) at every time slice. A history with $c(h)$ off-diagonal selections and $n - c(h)$ diagonal selections has exactly the product in (??). Factoring one $\cos(a\mu)$ from every slice gives (??). The Taylor statement is the standard expansion of the tangent. $\square$

The Lean development verifies the complete finite dictionary. Its recursive two-weight kernel satisfies

$$K[s, sw]_n = s^n K[1, w]_n,$$

and specialization to $s = \cos(a\mu)$ and $w = -i \tan(a\mu)$ gives the unitary checkerboard correspondence exactly when $\cos(a\mu) \neq 0$. A quarter-angle nondegenerate witness and the $a\mu = \pi/2$ pure-flip boundary are included. Separately, the development verifies that finite null histories weighted by a corner power equal a closed binomial kernel and satisfy a discrete Dirac recursion, and that the same direction-resolved history sum equals the corresponding transfer-matrix power. The direct Gaussian-rational path-sum declaration uses a real mass and corner weight $+i\epsilon m$ in its stated convention. Independently, the directed-edge theorem assigns every history edge the corresponding entry of the actual complex Plücker mass coin and proves that the complete oriented history sum equals its transfer-matrix power. Thus Ξ_z in (??) is now packaged directly: opposite turns carry z and $\bar{z}$, with the exact $3 + 4i$ control proving that the two normalized orientation factors are nonzero and unequal.

The tangent appearance is not asserted as novel by itself. Exponential and Euler-angle parametrizations of unitary Dirac-walk coins were related earlier, including explicit tangent formulas in quantum lattice-Boltzmann work [?]. The contribution here is the exact recursive-kernel scaling law composed with a coin whose generator is derived from the complex Plücker coordinate, together with its machine verification.

6 Full lattice spectrum, symmetry, and doubling audit

Quantum-walk claims cannot be judged only at small momentum. Introduce the dimensionless variables

$$q = ka \in [-\pi, \pi], \quad \theta = \mu a, \quad \Omega = \omega a.$$

The Brillouin zone is periodic in q , and quasienergy is periodic modulo 2π .

Theorem 6.1 (Complete two-band audit). *The two quasienergy bands are*

$$\Omega_{\pm}(q) = \pm \arccos(\cos q \cos \theta) \pmod{2\pi}. \quad (31)$$

They are degenerate only when $q \in \pi\mathbb{Z}$ and $\theta \in \pi\mathbb{Z}$. On the physical principal branch $0 < |\theta| < \pi/2$, the walk is gapped throughout the Brillouin zone: the positive rest energy at $q = 0$ is $|\theta|/a = \mu$, the interband zero-energy gap is 2μ , and the circular quasienergy gap at the zone edge is also $2|\theta|$. At $\theta = 0$ there are two massless cones:

$$(q, \Omega) = (0, 0) \quad \text{and} \quad (q, \Omega) = (\pi, \pi). \quad (32)$$

For a differentiable band away from a degeneracy,

$$v_g = \frac{\partial \omega}{\partial k} = \frac{\sin q \cos \theta}{\sin \Omega}, \quad |v_g| \leq 1, \quad (33)$$

with strict inequality for a massive angle whenever the denominator is nonzero.

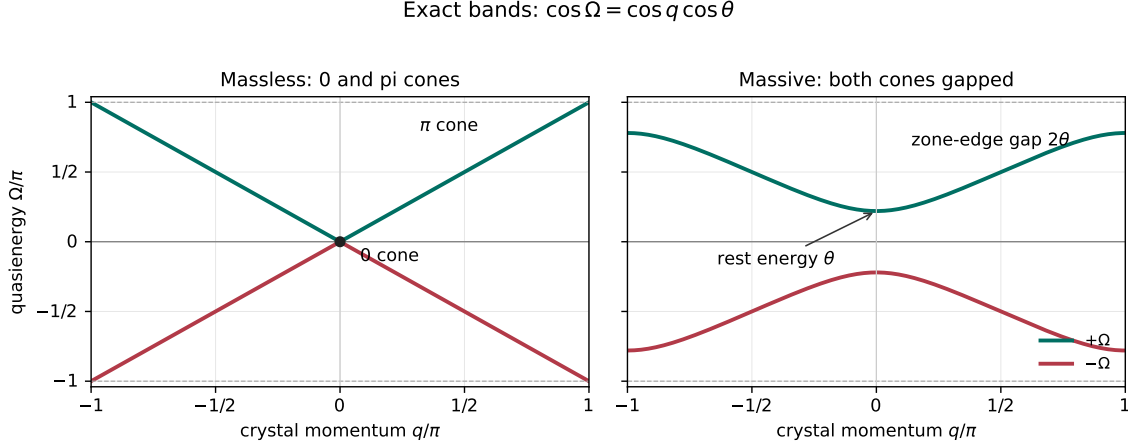


Figure 3: Exact quasienergy bands from $\cos \Omega = \cos q \cos \theta$. The left panel shows the massless cone at the origin and its Floquet partner at quasienergy π . A nonzero Plücker area opens both gaps (right). The curves evaluate the exact formula; they are not a fit or numerical simulation. Generated by `Scripts/sim/plot_null_spinor_dirac_bands.py`.

REALIZATION [KERNEL](#) for the trace-derived dispersion, velocity deficit inequality, exact massless set $\{0, \pi\}$, and uniform massive discriminant bound; [PROOF](#) for the implicit-differentiation identity in (??) and for packaging these facts into the displayed arccos branches.

Proof. Equation (??) is the principal solution of (??). A two-band degeneracy requires $\sin \Omega = 0$, equivalently $|\cos q \cos \theta| = 1$. Each factor has absolute value at most one, so equality forces $q, \theta \in \pi\mathbb{Z}$. At $q = 0$ one has $\Omega_+ = |\theta|$ on the stated branch; at $q = \pi$ one has $\Omega_+ = \pi - |\theta|$, whose circular distance to the negative band across the quasienergy boundary is $2|\theta|$. Implicit differentiation gives (??). Finally,

$$\sin^2 \Omega - \sin^2 q \cos^2 \theta = 1 - \cos^2 \theta = \sin^2 \theta \geq 0,$$

which proves the speed bound and its strict massive form. $\square$

Doubling and pseudo-doubling verdict. The high-momentum (π, π) cone is a Floquet zone-edge partner of the low-energy cone. It is not a second zero-quasienergy root, but it is a second relativistic band-touching point in the full lattice spectrum. Following the terminology of Gupta and Short [?], it is naturally classified as a pseudo-doubler: a high-quasienergy mode with a Dirac-like local dispersion. The elementary walk therefore does *not* establish a no-doubling or no-pseudo-doubling theorem. The bounded continuum estimate in Section ?? isolates fixed physical momenta as $a \rightarrow 0$, sending $q \simeq \pi$ to cutoff momentum; that decoupling is not the same as removing the finite-lattice partner.

Chiral time frame. Let

$$U_s(q, \theta) = e^{-i\theta\sigma_x/2} e^{-iq\sigma_z} e^{-i\theta\sigma_x/2}.$$

It is unitarily equivalent to $U(q, \theta)$ and obeys the exact chiral relation

$$\sigma_y U_s(q, \theta) \sigma_y = U_s(q, \theta)^\dagger. \quad (34)$$

Thus the reciprocal pairing of quasienergies is structural, not an artifact of the trace calculation. Equation (??) is the symmetry statement used here; a finer antiunitary or Floquet topological classification is left to a separate analysis [?].

Robustness. If z is perturbed to $z + \delta z$, then

$$||z + \delta z| - |z|| \leq |\delta z|.$$

Therefore every perturbation with $|\delta z| < |z|$ remains off the collinear gap-closing locus and retains a rest energy at least $|z| - |\delta z|$. The phase may rotate the internal basis, but by (??) it cannot change a constant-coefficient spectrum.

7 Uniform many-step Dirac control

The finite walk should approximate the continuous Dirac flow over a fixed macroscopic time, not merely at one infinitesimal step. For the real representative of the phase-conjugacy class, write

$$U(q, r) = e^{-iq\sigma_z} e^{-ir\sigma_x}, \quad H(k, \mu) = k\sigma_z + \mu\sigma_x, \quad V_t(k, \mu) = e^{-itH(k, \mu)}. \quad (35)$$

Theorem 7.1 (Uniform bounded-momentum product limit). *For $K, M \geq 0$, define*

$$C_{\text{box}}(K, M) = 2K^2 + 2M^2 + KM^2 + K^2M + KM, \quad (36)$$

$$D_{\text{box}}(K, M) = 4C_{\text{box}}(K, M) + 4(K + M)^2 e^{K+M}. \quad (37)$$

If $|k| \leq K$, $|\mu| \leq M$, $n > 0$, and $|t/n| \leq 1$, then

$$\left\| U\left(k\frac{t}{n}, \mu\frac{t}{n}\right)^n - V_t(k, \mu) \right\| \leq D_{\text{box}}(K, M) \frac{t^2}{n}. \quad (38)$$

The bound is uniform on the displayed parameter box, and its right-hand side tends to zero as $n \rightarrow \infty$.

NEW **KERNEL**

Proof architecture. The one-step walk differs from $\mathbf{1} - i(t/n)H$ by an explicit quadratic remainder. The exact exponential differs from the same first-order matrix by another quadratic remainder, giving $\|U - V_{t/n}\| \leq D(k, \mu)t^2/n^2$. Both matrices are unitary, so the power telescope has no norm-growth factor:

$$\|U^n - V_{t/n}^n\| \leq n \|U - V_{t/n}\|.$$

The exact short-time flows compose to V_t , producing the $1/n$ rate. Finally $D(k, \mu) \leq D_{\text{box}}(K, M)$ on the parameter box. Lean checks the entrywise remainder estimates, operator-norm conversion, unitarity, telescope, flow composition, box inequality, and limit. $\square$

This proof is a finite-dimensional first-order product-formula argument. Its constant is explicit rather than optimized; sharper general Trotter-error theory provides context for that distinction [?].

For $z = \mu u$ with $|u| = 1$, the diagonal unitary W_u commutes with Γ and conjugates B_μ to B_z . Unitary invariance of the operator norm therefore transfers (??) to the oriented Plücker walk. This one-line conjugacy corollary is **PROOF** pending a combined declaration.

The theorem proves uniform convergence of the finite momentum symbol in $1 + 1$ dimensions. It is not yet an infinite-volume inverse-Fourier theorem, a position-space PDE convergence theorem, or an interacting limit. Those quantifiers are separate problems, not implications hidden in the word “continuum.”

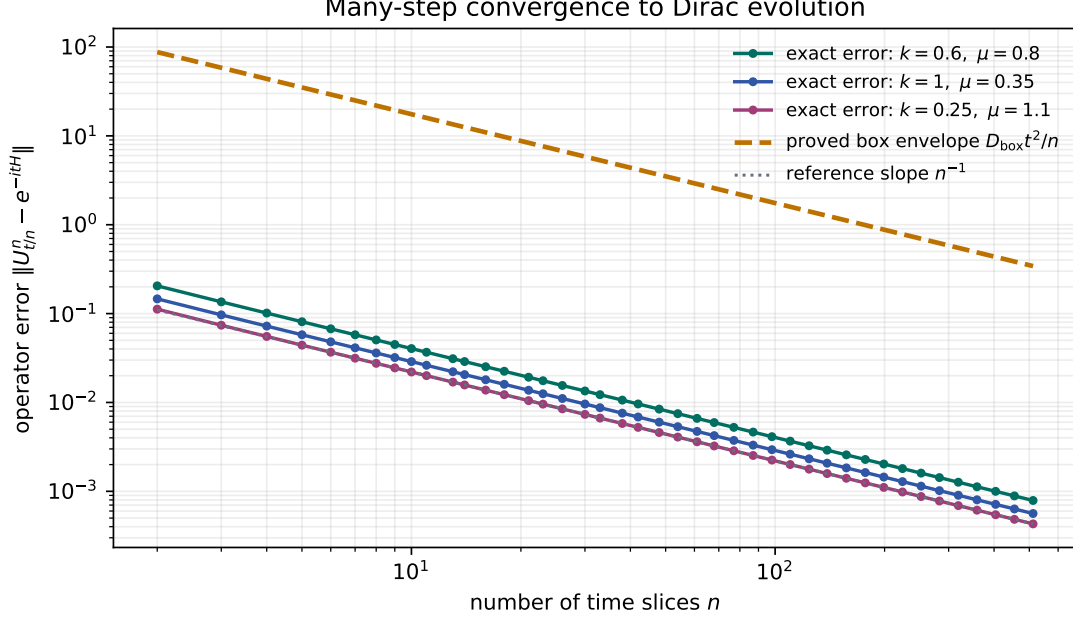


Figure 4: Exact many-step operator errors for three nondegenerate parameter choices. All follow the proved n^{-1} rate and remain below the explicit compact-box envelope. The points evaluate the exact finite matrices; the envelope and convergence statement are kernel checked.

8 Finite position kernels and a controlled $3 + 1$ extension

The uniform estimates already imply a finite position-space statement. Let $\mathcal{K}$ be a finite momentum set, let $\chi_x(k)$ be phases with $|\chi_x(k)| \leq 1$, and define the synthesized matrix kernel

$$\mathcal{S}_x[A] := \sum_{k \in \mathcal{K}} \chi_x(k) A(k).$$

Applying the triangle inequality to Theorem ?? gives the kernel-checked bound

$$\|\mathcal{S}_x[U_n] - \mathcal{S}_x[V_t]\| \leq |\mathcal{K}| D_{\text{box}}(K, M) \frac{t^2}{n}. \quad (39)$$

This controls every finite periodic Fourier kernel. The cardinality factor is the generic price of applying the triangle inequality pointwise. It disappears at the level of normalized L^2 wave packets. For Mathlib's unnormalized vector-valued transform $\mathcal{F}$ on $\mathbb{Z}/N\mathbb{Z}$, we prove

$$\sum_k \|(\mathcal{F}f)(k)\|^2 = N \sum_x \|f(x)\|^2, \quad \sum_x \|\mathcal{F}^{-1}g(x)\|^2 = \frac{1}{N} \sum_k \|g(k)\|^2. \quad (40)$$

Consequently, if a modewise error obeys $\|\delta(k)\| \leq \varepsilon \|c(k)\|$, then

$$\sum_x \|\mathcal{F}^{-1}\delta(x)\|^2 \leq \frac{\varepsilon^2}{N} \sum_k \|c(k)\|^2. \quad (41)$$

Both identities and the implication are `CLASSICAL KERNEL`; a one-mode packet is included as an exact normalization control. We also prove the operator-valued corollary: a uniform bound

$\|A(k) - B(k)\| \leq \varepsilon$ implies (??) for $\delta(k) = (A(k) - B(k))c(k)$. Applying that corollary to the local walk uses the exact plane-wave block stated below. Iterating the transform in the three coordinates gives, for $N \geq 1$, the kernel-checked product-torus identity

$$\sum_{x \in (\mathbb{Z}/N\mathbb{Z})^3} \|\mathcal{F}_3^{-1}g(x)\|^2 = \frac{1}{N^3} \sum_{k \in (\mathbb{Z}/N\mathbb{Z})^3} \|g(k)\|^2. \quad (42)$$

Hence, again for $N \geq 1$, a uniform modewise operator error ε gives the exact wave-packet coefficient ε^2/N^3 , with no cardinality loss. The momentum-sign bridge is also kernel-checked: for the positive-exponential convention the finite character block equals the analytic block at $k_j\varepsilon = -2\pi n_j/L$, including an exact quarter-zone $-i$ sign control.

There is also a countable successor to the finite Plancherel statement. If $e_n(k)$ is a countable family of mode errors such that $e_n(k) \rightarrow 0$ for every k and $\|e_n(k)\|^2 \leq b(k)$ for one summable envelope b , then

$$\sum_k \|e_n(k)\|^2 \rightarrow 0. \quad (43)$$

This is a direct vector-valued Tannery specialization, proved in Lean with a nonzero one-mode control. It isolates the remaining physical obligation: construct the walk-specific square-summable envelope and identify the limiting multiplier with the Dirac PDE propagator under a continuum Fourier isometry.

The corresponding measure-theoretic multiplier step is also closed. If $f \in L^2$ and pointwise errors satisfy $\|e_n(k)\| \leq \varepsilon_n \|f(k)\|$ almost everywhere, where $0 \leq \varepsilon_n$ and $\varepsilon_n \rightarrow 0$, then

$$\|e_n\|_{L^2} \rightarrow 0. \quad (44)$$

The proof is a kernel-checked ‘eLpNorm’ monotonicity-and-squeeze argument. The walk-specific compact-support application is also proved. For any fixed $t \in \mathbb{R}$, suppose $f \in L^2$ and, almost everywhere on the support of f , $|k_x|, |k_y|, |k_z| \leq K$ and $|z| \leq M$, with $K, M \geq 0$. Applying the complex split symbol for $n+1$ steps gives an error multiplier E_n satisfying

$$\|E_n f\|_{L^2} \rightarrow 0, \quad \|E_n(k)f(k)\| \leq \frac{D_{K,M}t^2}{n+1} \|f(k)\|. \quad (45)$$

The displayed pointwise envelope holds for all sufficiently large n ; the finite prefix is irrelevant to the sequence limit. This walk-specific composition is NEW **KERNEL**. What remains is the lattice/continuum Fourier-isometry and identification of the limiting multiplier with the position-space Dirac PDE, not compact-support multiplier convergence.

We now also close the changing-scale multiplier gate in $3+1$ dimensions. At refinement level N , set

$$h_N = \frac{1}{N+1}, \quad K_N = (N+1)^2, \quad W_N = 3(N+1) + M + 1, \quad qquad n_N = W_N^4. \quad (46)$$

The momentum grid spacing tends to zero, while its represented center radius is exactly $h_N K_N = N+1$ and therefore exhausts momentum space. For every integer mode k_N in the complete cutoff box and every fixed mass $|m| \leq M$, the live four-component split step obeys the uniform bound

$$\left\| U_{h_N k_N} \left(\frac{t}{n_N} \right)^{n_N} - e^{-itH(h_N k_N, m)} \right\| \leq \frac{2t^2}{W_N^2} \exp\left(\frac{|t|}{W_N^3} \right) \rightarrow 0. \quad (47)$$

This changing-spacing, exhausting-box theorem is NEW **KERNEL**. Its boundary fixture takes the positive x -face mode, whose represented physical momentum is exactly $N+1$; the result is therefore not a disguised fixed-compact-momentum limit.

A concrete cell-sampling dense core is also available. Let f be a complex Lipschitz momentum field with constant L , and let a finite family of the half-open cells of side h_N cover its support. Sampling f at each cell center gives the exact global estimate

$$\int_{\mathbb{R}^3} |I_N f(k) - f(k)|^2 dk \leq \#S_N h_N^3 \left(\frac{L h_N}{2} \right)^2. \quad (48)$$

Consequently, if the represented cell volume $\#S_N h_N^3$ is uniformly bounded and $h_N \rightarrow 0$, the left side tends to zero. Cell radius $h_N/2$ in the product sup norm, exact union volume, support reduction, and the limit are all NEW [KERNEL](#). This is strong $L^2(\mathbb{R}^3)$ convergence for a compact-support Lipschitz dense core. The regularity bridge to arbitrary data is also now kernel-checked: every complex $L^2(\mathbb{R}^3)$ field admits a compactly supported smooth approximant with arbitrarily small L^2 error, and that same approximant has one global Lipschitz constant. Compact support is load-bearing: the smooth noncompact quadratic $x \mapsto x_0^2$ has no global Lipschitz constant.

There is a sharp representative-level obstruction to extending this center sampler to arbitrary L^2 . A function equal to one at a single cell center and zero elsewhere is zero almost everywhere, but center sampling reads the value one; sampling the AE-equal zero function reads zero. Thus the center sampler is not AE-invariant and cannot define a bounded operation on L^2 equivalence classes. This no-go is NO-GO [KERNEL](#), not merely an analytic warning. Normalized cell averaging supplies the correct first repair: it is AE-invariant, sends the one-point spike to zero, and returns exactly one on the constant-one function for every positive cell size.

The representative-safe repair now closes the changing-lattice flow theorem for arbitrary square-integrable data. The finite cell-average projection is an exact L^2 contraction; compact support is eventually covered by the expanding schedule; only the active cells, not the full growing box, require a volume bound. The capstone then separates the total error into three terms: live evolution versus the exact flow at each cell center, exact flow at the center versus exact flow at the running momentum, and exact evolution of the cell-projection tail. Each term tends strongly to zero. Thus, for every fixed $z \neq 0$, fixed bound $M \geq |z|$, fixed time t , and fixed componentwise- L^2 four-spinor F ,

$$\|\mathcal{F}^{-1}(\text{live}_{N,z,t} P_N F - e^{-itH_z} F)\|_{L^2} \rightarrow 0. \quad (49)$$

Here the displayed live term abbreviates the cellwise reconstruction used in the formal statement, and $\mathcal{F}^{-1}$ is Mathlib's vector-valued inverse Fourier isometry. The representative identity, all three norm limits, their triangle composition, and the position-space norm transfer are NEW [KERNEL](#). This is a changing-lattice, full- L^2 convergence theorem for fixed mass, data, and time; it is not uniform over all masses or data, and it says nothing about interactions or the extra high-frequency sectors audited below.

The exact comparison flow also has its expected generator on compact momentum support: the strong derivative at $t = 0$ is pointwise $-iH_z(q)F(q)$. Independently, the affine Hermitian symbol $H_z(q)$ now defines a densely defined maximal multiplication operator on momentum-space L^2 . Explicit bounded inverses for $H_z \pm i$ have operator norm at most one, map into the maximal graph domain, and make both imaginary shifts surjective. The resulting unbounded Hamiltonian is therefore self-adjoint, equal to its adjoint, and closed; these statements are NEW [KERNEL](#).

Two analytic identifications remain open and are not hidden in (??). First, the Fourier conjugate of this maximal multiplier has not yet been proved equal to the position-space differential Dirac operator with Mathlib's exact 2π , sign, and i convention. Second, its inverse-Fourier graph domain has not yet been identified with a vector-valued Sobolev H^1 domain. Convergence of derivatives and interacting dynamics remain later gates as well.

The same architecture extends to four internal components. The successive-axis factorization itself is the standard construction used by Mlodinow and Brun [?]; what is new here is the Plücker-derived complex coin, the exact finite-torus implementation with a two-sided inverse and strict causal cone, and the kernel-checked all-zone audit that follows. Let Hermitian matrices $\alpha_1, \alpha_2, \alpha_3, \beta \in M_4(\mathbb{C})$ satisfy

$$\alpha_i^2 = \beta^2 = \mathbf{1}, \quad \{\alpha_i, \alpha_j\} = 0 \ (i \neq j), \quad \{\alpha_i, \beta\} = 0,$$

and set

$$H_4(\mathbf{k}, m) = \sum_{j=1}^3 k_j \alpha_j + m \beta.$$

The explicit matrices used in the artifact obey

$$H_4(\mathbf{k}, m)^2 = (|\mathbf{k}|^2 + m^2) \mathbf{1}. \quad (50)$$

For a substep ε , define

$$U_\varepsilon^{(4)}(\mathbf{k}, m) = e^{-i\varepsilon k_x \alpha_1} e^{-i\varepsilon k_y \alpha_2} e^{-i\varepsilon k_z \alpha_3} e^{-i\varepsilon m \beta}. \quad (51)$$

The order in (??) is not cosmetic: the spatial generators do not commute, and the artifact contains that noncommutation as a negative control against a false second-order claim.

Theorem 8.1 (Finite local 3 + 1 walk and compact rate). *On a finite periodic three-dimensional position register, there is an explicit successive-axis update built from one-site conditional shifts and the mass coin in (??). It preserves the finite inner product exactly, has an explicit two-sided inverse, and reads only sites reached by one conditional source shift along each axis in one cycle. Its product plane-wave sectors are invariant, and on each sector the update is exactly the ordered axis-0/axis-1/axis-2/mass matrix block built from finite roots of unity. For the positive-exponential plane-wave convention, that block is (??) when*

$$k_j \varepsilon = -\frac{2\pi n_j}{L}. \quad (52)$$

If $|k_x|, |k_y|, |k_z| \leq K$, $|m| \leq M$, $n > 0$, and $|t/n| \leq 1$, then

$$\left\| \left(U_{t/n}^{(4)}(\mathbf{k}, m) \right)^n - e^{-itH_4(\mathbf{k}, m)} \right\| \leq 16(3K + M)^2 e^{3K+M} \frac{t^2}{n}. \quad (53)$$

Every finite Fourier synthesis obeys the same bound multiplied by its number of modes, while the normalized product-torus L^2 wave-packet error obeys the squared bound divided by N^3 . Moreover, let $B_4(z)$ be the phase-retaining rest operator and let $U_\varepsilon^{(4)}(\mathbf{k}, z)$ be the chiral conjugate of the displayed step at $m = |z|$. Its exact comparison flow is generated by

$$H_4(\mathbf{k}, z) = \sum_{j=1}^3 k_j \alpha_j + B_4(z), \quad H_4(\mathbf{k}, z)^2 = (|\mathbf{k}|^2 + |z|^2) \mathbf{1},$$

and whenever $|z| \leq M$ it satisfies the same bound

$$\left\| \left(U_{t/n}^{(4)}(\mathbf{k}, z) \right)^n - e^{-itH_4(\mathbf{k}, z)} \right\| \leq 16(3K + M)^2 e^{3K+M} \frac{t^2}{n}. \quad (54)$$

REALIZATION **KERNEL** for exact local norm preservation, Clifford square, finite-character plane-wave block, compact rate, finite synthesis bound, and the trigonometric/sign conversion (??) to the analytic symbol; NEW **KERNEL** for the complex-phase transfer (??) and the iterated three-torus normalization (??).

Proof architecture. Each α_j has eigenvalues ± 1 . An explicit unitary eigenbasis turns the corresponding diagonal one-site conditional shift into the physical Clifford factor; composing those shifts with the mass coin proves exact norm preservation. Direct substitution into every product plane wave extracts one unit-modulus character per conditional shift and proves the exact ordered finite block without an asymptotic argument. The closed form $e^{-iq\alpha_j} = \cos q \mathbf{1} - i \sin q \alpha_j$ identifies the ordered analytic matrix symbol after the sign convention in (??). A four-factor quadratic remainder estimate, exact unitarity, and the unitary power telescope then give (??). Applying the finite synthesis theorem gives its position-kernel corollary. $\square$

For $z \neq 0$, the phase transfer is exact. If W_z is the chiral unitary at the polar angle of z , then

$$W_z H_4(\mathbf{k}, |z|) W_z^\dagger = H_4(\mathbf{k}, z), \quad W_z e^{-itH_4(\mathbf{k}, |z|)} W_z^\dagger = e^{-itH_4(\mathbf{k}, z)}.$$

The operator norm cannot increase under this conjugation, so the real-mass estimate transfers without changing its constant.

The 3 + 1 high-symmetry verdict

The continuum tangent is exactly isotropic: $H_4(\mathbf{k}, 0)^2 = |\mathbf{k}|^2 \mathbf{1}$. The finite cubic walk is not globally doubler-free. At the eight corners $(q_x, q_y, q_z) \in \{0, \pi\}^3$, its massless symbol is

$$U^{(4)}(q_x, q_y, q_z, 0) = (-1)^r \mathbf{1}, \quad r = \#\{j : q_j = \pi\}. \quad (55)$$

Thus $(\pi, \pi, 0)$, $(\pi, 0, \pi)$, and $(0, \pi, \pi)$ are distinct momenta with exactly the same zero-quasienergy symbol as the origin. Equation (??), all three distinctness witnesses, and radial isotropy of the continuum square are REALIZATION **KERNEL**. This is an actual finite-regulator obstruction, not a caveat inferred from a plot.

The massive walk has a stronger interior obstruction. At body-center momentum $\mathbf{q}_B = (\pi/2, \pi/2, \pi/2)$, let $u_\theta = \cos \theta + i \sin \theta$ and

$$v_+(\theta) = (-u_\theta, 0, 1, 0)^T, \quad v_-(\theta) = (u_\theta, 0, 1, 0)^T.$$

For every real mass angle θ , the exact finite step obeys

$$U^{(4)}(\mathbf{q}_B, \theta) v_+(\theta) = v_+(\theta), \quad U^{(4)}(\mathbf{q}_B, \theta) v_-(\theta) = -v_-(\theta), \quad v_\pm(\theta) \neq 0. \quad (56)$$

Equivalently, the body-center step is a non-scalar involution for all θ . It therefore retains exact quasienergies 0 and π even after the mass coin is turned on: the present cubic regulator admits no uniform Floquet gap (a body-center crossing at 0 and π persists for every mass angle). The closed body-center matrix, involution identity, and both explicit eigenmodes are NO-GO **KERNEL**.

The interior crossings moreover carry an exact charge bookkeeping. At each of the eight nodes $q_j \in \{\pi/2, 3\pi/2\}$ the massive step has two-dimensional $+1$ and -1 eigenspaces simultaneously, and the Schur-reduced crossing Jacobian of every node and gap is one of exactly two diagonal matrices, $\pm \frac{4}{5} \text{diag}(-1, 1, -1)$, with determinant $\pm(\cos \theta)^3$ at the 3-4-5 angle — every node is a nondegenerate cone. The resulting local charges (the Jacobian determinant signs) are distributed by the parity of the node's $3\pi/2$ -coordinates, are opposite at the two quasienergies of each node

(the Floquet pairing), and sum to zero at each quasienergy separately. This is now machine-checked from the walk symbol itself. The base census arithmetic — the two Jacobian determinants, the parity-indexed assignment, and its sum-zero — is kernel-checked in `SplitStepChargeBalance`; and the derivation of the Jacobians *from the Bloch symbol* is kernel-checked for all eight nodes in `SplitStepSchurJetAllNodes`: the exact node collapse $U_n = (-1)^{\text{par}(n)}U_0$ (from $\cos \frac{3\pi}{2} = 0$, $\sin \frac{3\pi}{2} = -1$), the two-dimensional ± 1 kernels at every node, the Schur-reduced 3×3 Jacobian, and the resulting per-node Floquet opposition and sum-zero. The derivation is anchored to the landed census by a compiler-enforced bridge (`CensusDerivationBridge`): the walk-derived assignment is proved equal, at every node and gap, to the `SplitStepChargeBalance` census imported from that module, so nothing rests on an unchecked copy. The one modelling choice made explicit is that the nondegenerate-cone reading uses the normal-matrix structure of each node. The doublers of this walk are not merely present: they are exactly the charge partners the sum rule demands — derived from the symbol — and the two gaps compensate node by node.

The obstruction is also visible in an exact all-zone criterion. Write $x = \cos q_x$, $y = \cos q_y$, $w = \cos q_z$, and $c = \cos \theta$, and define

$$S = 4c^2x^2y^2w^2 - 2c^2x^2y^2 - 2c^2x^2w^2 + c^2x^2 - 2c^2y^2w^2 + c^2y^2 + c^2w^2 - 2x^2y^2w^2 + x^2y^2 + x^2w^2 + y^2w^2, \quad (57)$$

$$P_0 = S - 2cxyw, \quad P_\pi = S + 2cxyw. \quad (58)$$

For the live ordered successive-axis symbol, the finite determinant expansion is now exact at every real momentum and mass angle:

$$\det(U^{(4)} - \mathbf{1}) = 4P_0, \quad \det(U^{(4)} + \mathbf{1}) = 4P_\pi. \quad (59)$$

Thus $P_0 = 0$ and $P_\pi = 0$ are respectively necessary and sufficient for zero- and π -quasienergy modes. At body center $x = y = w = 0$, both polynomials vanish for every θ , recovering (??). The two determinant identities, their explicit 4×4 expansion, and the definitional bridge to the walk used in the continuum estimate are NO-GO `KERNEL`.

The zero sets are now classified on the principal massive branch. If $0 < |c| < 1$ and $|x|, |y|, |w| \leq 1$, then

$$P_0 = 0 \iff x = y = w = 0 \iff P_\pi = 0. \quad (60)$$

The proof uses the exact decomposition

$$P_0 = (cx - yw)^2 + (y^2 + w^2 - 2y^2w^2)(c^2 + x^2(1 - 2c^2)), \quad (61)$$

whose second factor is strictly positive and whose first prefactor is nonnegative on this branch. Equality then forces all three momentum cosines to vanish; the P_π statement follows by $c \mapsto -c$. Thus the simultaneous body-center cosine locus is the complete zero- and π -quasienergy crossing set for every principal massive angle, not merely a sampled obstruction. The hypotheses are load-bearing: at $c = 0$, additional zeros occur, including $(x, y, w) = (1, 0, 0)$. The classification and this boundary control are NO-GO `KERNEL`. They classify the failure of the present regulator; they do not make it alias-free.

A degree-one stationary-amplitude no-go. The most economical proposed repair is to add a stay-put channel to each nearest-neighbor factor. The exact ansatz is

$$F(q) = e^{iq}A + B + e^{-iq}C. \quad (62)$$

Let M be any Hermitian involution. If $F(q)$ is unitary for every real q , $F(0) = \mathbf{1}$, and $F'(0) = -iM$, then

$$B = 0. \quad (63)$$

The proof evaluates the exact unitarity residual at $q = \pm\pi/2$; adding the two resulting matrix identities forces $B^\dagger B = 0$. Specializing $M = \alpha_j$ rules out a nonzero stationary amplitude in a degree-one factor with the full Dirac tangent, even before zone-edge separation is imposed. All three specializations to the live $3 + 1$ generators are separately kernel-checked. This is NO-GO [KERNEL](#). A nonvacuity control shows that a partial-channel Hermitian tangent with a stationary kernel does admit an exactly unitary degree-one witness with $B \neq 0$ and a non-scalar zone edge. That witness does not retain the full Dirac tangent. Thus stationary amplitudes remain viable only after a genuine change of architecture, such as a larger internal cell, longer Laurent range, multistep factorization, or a non-involutory tangent.

The last route is not hypothetical. Gupta and Short [?] construct doubler- and pseudo-doubler-free nearest-neighbor one-step unitary walks, in $1 + 1$ and $3 + 1$ dimensions, whose enabling resource is exactly a nonzero stay-put amplitude $\gamma_0 = \mathbf{1} - \gamma_+ - \gamma_-$ built from tilted, non-orthogonal velocity projectors (their Eqs. (29)–(30) and (37)). Their step is exactly of the degree-one Laurent form (??), unitary at every momentum and normalized at the origin, with stationary block γ_0 . The no-go (??) therefore yields a corollary about their family: its tangent generator, though automatically Hermitian for a unitary origin-normalized step, cannot be an involution ($M^2 \neq \mathbf{1}$). The stay-put freedom that removes the doublers is purchased exactly by relinquishing the involutory unit-speed Dirac tangent, and our partial-channel escape witness realizes the same mechanism in kernel-checked form. Their $3 + 1$ family moreover retains extraneous Weyl-like low-energy solutions at isolated momenta $\pm \mathbf{q}(\theta)$, which the authors suggest a more general higher-dimensional construction might remove (their Appendix F), so a strictly local, exactly unitary, fully alias-free $3 + 1$ walk with the complete involutory Dirac tangent remains open in both programs.

The obstruction is stronger for a separable three-axis product. Once $B_x = B_y = B_z = 0$, origin normalization forces each factor to obey $F_j(\pi) = -\mathbf{1}$. Therefore, for any momentum-independent onsite coin Q , including the complex Plücker mass coin,

$$U_Q(0, 0, 0) = U_Q(\pi, \pi, 0) = U_Q(\pi, 0, \pi) = U_Q(0, \pi, \pi). \quad (64)$$

This all-coins alias theorem is NO-GO [KERNEL](#). It proves that changing the onsite mass data cannot repair the cubic partners inside the four-channel, range-one, single-factor-per-axis architecture. Extra range, coupled substeps, or enlarged/ancillary cells are not optional embellishments; at least one is structurally required.

An attempted commutator–Wilson repair sharpens that requirement. Consider the exactly unitary one-axis gate

$$W_r(q) = F_\beta(rq)F_{\gamma_5}(rq)F_\beta(-rq)F_{\gamma_5}(-rq), \quad F_A(s) = \cos s \mathbf{1} - i \sin s A. \quad (65)$$

Its linear term cancels, so at the Bloch-matrix level it appears to supply a second-order Wilson-like correction without disturbing the Dirac tangent. However, in the fixed one-cell ansatz strict finite Laurent range requires the frequency r to be an integer. For every integer r the exact zone-edge identity is

$$W_r(\pi) = \mathbf{1}, \quad (66)$$

and all three even-corner aliases in (??) remain exactly equal to the origin. Conversely, the noninteger trigonometric values that lift selected corners are not 2π -periodic finite Laurent polynomials in e^{iq}

and therefore do not define the claimed strict local one-cell QCA. This family-scoped locality/dealiasing incompatibility, including the range-one control, is NO-GO **KERNEL**. It does not exclude a larger cell or a genuinely different finite product; it prevents a nonlocal Bloch control from being mislabeled as that successor.

The matching cross-architecture algebraic invariant is now exact in one Laurent variable. The determinant of every invertible finite Laurent matrix is a nonzero scalar times a uniquely exponented monomial, and that exponent is additive under composition, vanishes on the identity, and equals n on the one-channel shift T^n . This is the algebraic precursor of the translation-invariant one-particle quantum-walk index in GNVW Proposition 5 [?]; identifying the two still requires a complex-unitary Laurent/Fourier representation theorem and an explicit shift-sign convention. It is not the positive-rational many-body GNVW QCA index, an entropy-flow theorem, or a no-doubling result.

There is also a necessary correction to the charge strategy. A complete mass-admitting four-component Dirac crossing admits an explicit gapped algebraic contraction to a constant. In the standard class-A topological reading this makes the complete crossing neutral, but that topological bridge is not itself part of the kernel result. If $F_0(\mathbf{n}) = \sum_j n_j \alpha_j$ is the flattened tangent and β anticommutes with it, then

$$F_s(\mathbf{n}) = \cos(\pi s/2)F_0(\mathbf{n}) + \sin(\pi s/2)\beta \quad (67)$$

satisfies $F_s(\mathbf{n})^2 = \mathbf{1}$ and ends at the constant matrix β . The finite matrix identity, including explicit anticommuting involutions and a missing-anticommutation control, is REALIZATION **KERNEL**. The nonzero local charges instead belong to the two Weyl sectors selected by $\Xi = -i\alpha_1\alpha_2\alpha_3$. For the actual live cubic Clifford tangent, an exact finite calculation constructs two orthogonal Weyl bases and restricts $(\alpha_1, \alpha_2, \alpha_3)$ to the Pauli triple and its negative. The resulting supplied-Jacobian charges are exactly $+1$ and -1 , so their four-component sum vanishes; a singular tangent has charge zero. This is REALIZATION **KERNEL** at the local tangent. A sectorwise Brillouin-zone argument still requires the additional global hypothesis $[U(\mathbf{q}), \Xi] = 0$ at every momentum; local chirality at the origin is insufficient. For the actual ordered Bloch step this gate is now exact:

$$[U(\mathbf{q}, \theta), \Xi] = 0 \quad \Longleftrightarrow \quad \sin \theta = 0. \quad (68)$$

The chirality is a Hermitian involution, and its positive-sector projector commutes with the complete massless walk at every momentum. The walk is therefore globally Weyl-sector split, while every genuine mass angle ($\sin \theta \neq 0$) mixes the sectors; an exact quarter-turn fixture gives a nonzero commutator. Here the mass weight is $\sin \theta$; a cosine variable used in the separate determinant factorization is not that weight. These statements are NEW **KERNEL**. They determine precisely when the sectorwise no-go can be invoked, but it does not supply the missing global charge-sum theorem.

The analytic and arithmetic core of that census is now machine-checked on the 2×2 positive-Weyl block. Lean extracts the exact Pauli vector of the live ordered step, proves its complete real Fréchet derivative, and identifies the derivative matrix with the one used in the finite census. Its determinant obeys

$$\det J(\mathbf{q}) = u_0(\mathbf{q})(\cos^2 q_y - \sin^2 q_y), \quad (69)$$

evaluates to ± 1 at the sixteen designated principal-torus nodes, and the actual live-Jacobian charges cancel separately on the displayed $U = +1$ and $U = -1$ lists. Every designated node consequently has a distinct nondegenerate partner inside its finite list. These are NEW **KERNEL**. A second exact theorem classifies the complete massless 4×4 zero- and π -determinant sets in cosine coordinates: each is the appropriate parity class of cube corners together with the body-center locus. The

branch-sensitive bridge is now exact. For the actual positive 2×2 Weyl block, $U = +\mathbf{1}$ holds if and only if either all three cosine squares are one with cosine product $+1$, or all three cosines vanish with sine product -1 . Likewise, $U = -\mathbf{1}$ holds exactly on the odd-parity cosine corners or the body centers with sine product $+1$. These two global trigonometric loci are disjoint, and the actual Fréchet-Jacobian determinant is nonzero at every point of either locus. The rank-deficient quarter-axis control lies on neither branch. This is NEW [KERNEL](#) and upgrades the displayed charge census from a selected list to an exhaustive positive-Weyl branch classification. It is still a Jacobian-sign doubling theorem, not a formal Chern-number computation. At a full 4×4 body center both quasienergies remain present across the two Weyl blocks, which is why the branch theorem could not be inferred from the full determinant set alone.

This distinction sharpens both the no-go and the escape. Read's algebraic K -theory calculation for complex Laurent automorphisms retains only the constant unit and coordinate-delay classes, with no higher-dimensional strong component in the change-of-rings image [?]. Together with Floquet charge–bulk bookkeeping [?], this supplies a source-supported route to a global-chirality doubling theorem, but the finite-rank and zero/ π convention composition is not yet kernel-checked here. Conversely, the exact decomposition $U_{\perp} = (U - \Xi U \Xi)/2$ proves that $U_{\perp}$ has zero constant and linear coefficients when the normalized Dirac first jet commutes with Ξ . Therefore the first construction class worth testing consists of exact Laurent-unitary regulators with nonzero chirality mixing beginning beyond the linear jet, not another globally chirality-preserving onsite coin.

That escape resource is now exact at coefficient level. The quadratic regulator $q(\mathbf{k})R$, with $q(\mathbf{k}) = \sum_j k_j^2$ and an explicit chirality-odd matrix R , vanishes at the Dirac point and has zero complete Fréchet derivative there, yet equals a nonzero chirality-mixing matrix on a unit-axis fixture. This is NEW [KERNEL](#). It proves that higher-order mixing can preserve the desired first jet; it does not yet make the regulator an exact finite-range unitary or exclude every unwanted crossing.

The compatible exact-unitary primitive is a four-factor group commutator of circle phase steps. Kernel-checked, it is unitary, becomes the identity when either angle vanishes, gives $+\mathbf{1}$ for commuting involutions and $-\mathbf{1}$ for anticommuting involutions at a quarter turn, and admits an explicit rational mixed pair whose result is not any scalar matrix. Its algebraic Lie coefficient $GA - AG$ is chirality-odd whenever A is chirality-even and G is chirality-odd; for the live (α_1, β) pair that coefficient is nonzero and equals its full odd projection. The complete first Fréchet derivative of the commutator vanishes at the origin, while its mixed second derivative is exactly $GA - AG$. These are NEW [KERNEL](#). The same exact algebra also proves a sharp negative result. A common sign flip of either cosine/sine pair leaves the commutator unchanged because the corresponding central signs occur twice and cancel. Hence every integer-frequency phase commutator, including one with affine phase offsets, takes the same value at all cubic $0/\pi$ corners as at the origin; every finite product of zero-offset loops is exactly invisible there. A single phase step supplies the negative control and does see the sign flip. Thus the obstruction belongs to the doubled commutator architecture, not to phase gates individually.

A construction that survives that obstruction uses a genuinely noncentral conditional shift. With $D(z) = \text{diag}(z, 1)$, an exact rational coin C , and $K(z) = D(z)CD(z^{-1})C^{-1}$, define the reciprocal word $S(z) = K(z)K(z^{-1})$. Lean proves that S is unitary on the circle, $\det S = 1$, $S(1) = \mathbf{1}$, and $S(z) - \mathbf{1} = (z - 1)^2 Q(z)$. Thus its constant and linear jets vanish exactly. Its strict locality is also exact rather than inferred: S has five Laurent coefficients at powers $-2, -1, 0, 1, 2$, both range-two coefficients are nonzero, their zeroth moment is the identity, and their first moment vanishes. After multiplication by z^2 , the word is an ordinary degree-four matrix polynomial. At $z = -1$ the resulting rational matrix has both $\det(S - \mathbf{1}) = 2304/625$ and $\det(S + \mathbf{1}) = 196/625$, so it removes both zero- and π -crossings of the identity baseline at that corner. This is NEW [KERNEL](#) as a two-band finite-Laurent construction primitive, not yet an alias-free $3 + 1$ walk. A naive four-component

embedding creates additional generic roots in exact-arithmetic tests; any successful embedding owes an exact all-torus root certificate.

The two-band primitive itself is now classified across the full rational tangent-half-angle coin family. Writing $c = (1 - r^2)/(1 + r^2)$ and $s = 2r/(1 + r^2)$, the reciprocal word remains exactly unitary with determinant one on the circle. At $z = -1$, its zero-gap determinant is nonzero exactly when $r \neq 0, \pm 1$, while its π -gap determinant is nonzero exactly away from the roots of $r^2 - 2r - 1$ and $r^2 + 2r - 1$. The original 3–4–5 coin is exactly the $r = 1/2$ member, and $r = 0$ is the identity negative control. These are NEW **KERNEL** two-band corner statements. They do not define the four-component blocks or provide a global-torus embedding certificate.

There is a precise gate on one tempting certificate. For a unitary 4×4 step with $\det U = 1$, the determinant $\det(U - \mathbf{1})$ is exactly real: the kernel-checked identity is

$$\overline{\det(U - \mathbf{1})} = \det(U - \mathbf{1}).$$

This makes a real intermediate-value argument available when two Weyl sectors have paired determinant. Conversely, real endpoint values alone do not make $\det(U(q) - \mathbf{1})$ real along a path. The family-level reciprocal sign oracle therefore remains conditional until the live $U_{\pm}(q)$ blocks, one consistent register convention, paired determinant along an explicit origin-avoiding path, and the endpoint formulas are composed in one theorem. This condition is a kernel-checked reusable reality gate, not a root-existence theorem.

At the two-band level there is a complementary exact locking theorem. If U is unitary and $\det U = 1$, then

$$\det(U - \mathbf{1}) = 0 \iff U = \mathbf{1}, \quad \det(U + \mathbf{1}) = 0 \iff U = -\mathbf{1}.$$

Thus determinant roots and whole-sector zero/ π crossings coincide in an $SU(2)$ block; they are not merely statements that one eigenvalue happened to hit the branch. Exact determinant-minus-one controls show the conclusion fails without unimodularity. This is NEW **KERNEL** and supplies the legal bridge between matrix-root and identity-preimage censuses used below.

A more serious escape attempt puts the reciprocal mixing inside a genuinely chirality-coupled four-component coin, so that the regulator is noncentral at the old corners and leaves the Dirac first jet only at quadratic order. That direct construction also fails. On the exact slice $q_x = \pi$, $q_y = 0$, its zero- and π -crossing determinants reduce to squared reciprocal quartics. With $x = z + z^{-1} = 2 \cos q_z$, their reduced factors have opposite signs on $x \in [1, 2]$ and $x \in [-1, 0]$, respectively. Hence the coupled word has at least one nonzero-momentum crossing of each quasienergy on that slice. Lean proves both physical unit-circle existentials exactly: the reduced real quadratics cross zero, the intermediate value theorem supplies roots, and q_z is reconstructed with arccos, with $q_z \neq 0$ in both cases. This is NO-GO **KERNEL** for the displayed direct construction, not a universal reciprocal no-go. The negative result is diagnostic: corner visibility and higher-order chirality mixing can move the alias locus without reducing it. An enlarged determinant-paired register or an explicitly minimally doubled construction is the next construction class to test.

A second exact primitive shows that a stationary onsite amplitude need not conflict with strict locality or unitarity. For arbitrary orthogonal projectors P and Q , not assumed to commute, the controlled-phase product has the range-one Laurent form

$$U(z) = z\Gamma_+ + \Gamma_0 + z^{-1}\Gamma_-, \quad |z| = 1,$$

and is exactly unitary. A rational noncommuting two-band witness has $\Gamma_0 = \begin{pmatrix} 16/25 & -12/25 \\ 12/25 & 16/25 \end{pmatrix} \neq 0$. The necessary $z \neq 0$ hypothesis and the correct same-family adjoint laws were enforced during

formalization after the originally proposed formulas failed. This is NEW [KERNEL](#) as a one-variable construction mechanism inspired by the stationary-amplitude route of Gupta and Short. It does not yet supply a three-dimensional isotropic symbol, a Dirac tangent, or an all-zone proof that only the intended Weyl cones remain.

One of those gaps has a constructive answer. Three explicit rational noncommuting projector pairs have first Laurent moments exactly $(3/5)\sigma_x$, $(3/5)\sigma_y$, and $(3/5)\sigma_z$. Every axis retains a nonzero onsite amplitude, while their ordered three-axis product is exactly unitary for every torus momentum and equals the identity at the origin. Thus strict locality, stationary amplitude, all-zone unitarity, and an isotropic set of per-axis Weyl first moments are compatible in one NEW [KERNEL](#) construction. At the origin, the other two factors are identities, so the product-level isotropic first jet follows by the ordinary product rule; that combined derivative has not yet been packaged as a separate kernel theorem. The claim stops there: the full zero- and π -crossing census of this new symbol must determine whether it is minimally doubled or still more highly aliased.

A global test is negative. At the distinct torus phase $(z_x, z_y, z_z) = (-1, 1, -1)$, the complete ordered symbol is exactly the identity. Both the matrix equality and distinctness from $(1, 1, 1)$ are NO-GO [KERNEL](#). Thus the explicit stationary-amplitude construction is not unique-cone. This is not the hollow case of a constant identity symbol: at $(z_x, z_y, z_z) = (i, 1, 1)$ its $(0, 0)$ entry is exactly $16/25$, so the symbol is kernel-proved nonconstant. Exact-coefficient computation identifies two additional off-corner identity roots. One is the following kernel theorem:

$$(z_x, z_y, z_z) = \left(\frac{-9 - 40i}{41}, \frac{-9 + 40i}{41}, 1 \right), \quad U(z_x, z_y, z_z) = I.$$

Both nontrivial phases lie exactly on the unit circle and differ from 1. Conjugate pairing is load-bearing: replacing z_y by z_x makes the $(0, 1)$ matrix entry exactly $-1104/1681$, so the resulting symbol is not the identity. This off-corner crossing is NO-GO [KERNEL](#). A second, fully off-axis crossing is exact as well. In tangent coordinates its third coordinate is the unique real root of

$$480t^5 - 575t^4 - 1026t^2 + 1440t - 575 = 0, \quad 149/100 < t < 3/2.$$

Explicit rational quartic functions of t reconstruct the other two tangent coordinates. Lean proves all three are nonzero, maps them to unit-circle phases different from 1, and proves that the actual imported live matrix is exactly $U = I$. The companion elimination sextic is strictly positive on all of $\mathbb{R}$. The corrected original-generator certificate is now exact in Lean as well:

$$(1 + t_z^2)^2 t_z p_5(t_z) p_6(t_z) = Q_x F_x + Q_y F_y + Q_z F_z.$$

The three quotient polynomials contain 522 exact rational terms in total. Kernel normalization proves this identity and, because $1 + t_z^2 > 0$, proves that simultaneous numerator zeros force the zero, quintic, or sextic branch. The chart factor is mandatory: the corresponding bare polynomial is not in the numerator ideal. These are NEW [KERNEL](#). The formerly missing live bridge is now exact: on the finite real tangent chart, Lean proves that the actual imported matrix satisfies $U = I$ if and only if its scalar equation and all three numerator equations hold. On the full $t_z = 0$ slice, those three live numerator equations have exactly two real solutions, the origin and $(-5/4, 5/4, 0)$; both reconstruct $U = I$, and the latter is precisely the 9–40–41 alias. The interior census is now complete: exact polynomial-ideal certificates force t_x and t_y to be the displayed quartic reconstruction functions of t_z on the unique real quintic branch. Hence every finite-chart identity crossing is exactly the origin, the 9–40–41 point, or the unique fibre over that quintic root; the last branch has a fully off-axis witness. NEW [KERNEL](#) An exact boundary oracle finds only the already known $(-1, 1, -1)$ identity crossing on the seven phase-1 chart strata, but that boundary classification remains a

separate formal gate. The result is therefore an exhaustive finite-chart no-go with a boundary-chart obligation, not a finished regulator.

A finite $\mathbb{Z}_2^3$ cover can fold the eight $\{0, \pi\}^3$ corner labels to one reduced representative while retaining a unique three-bit flavour label. The cover is bijective, its deck action is free and transitive, and a single diagonal $\mathbb{Z}_2$ bit is provably too small to label all corners. This is NEW **KERNEL** as an exact census, but not a cure: it relabels eight crossings and removes none. For the present successive-axis symbol the exact translation intertwiner reduces to a scalar parity sign. Four flavour labels preserve zero quasienergy and four exchange zero with pi; a one-axis translation is explicitly nonidentity, while a pi/2 shift fails to be any scalar deck action. These statements are NEW **KERNEL**. This route must therefore not be advertised as a unique-cone theorem or as a derivation of observed particle generations: it is a faithful eightfold census.

The finite logical hinge is also kernel-checked: if a finite crossing set has total integer Jacobian-sign charge zero and one crossing is nondegenerate, a distinct nondegenerate crossing must exist. The theorem makes the topological input visible rather than hiding it: the zero-total premise still has to come from an exact crossing census or the separately audited Laurent/Floquet composition.

There is already a constructive Hamiltonian-level route around this no-go. Using the same live Clifford generators, define

$$M_W(\mathbf{q}) = m + r \sum_{j=1}^3 (1 - \cos q_j), \quad (70)$$

$$H_W(\mathbf{q}) = \sum_{j=1}^3 \sin q_j \alpha_j + M_W(\mathbf{q}) \beta. \quad (71)$$

The kernel-checked Wilson identity is

$$H_W(\mathbf{q})^2 = \left(\sum_{j=1}^3 \sin^2 q_j + M_W(\mathbf{q})^2 \right) \mathbf{1}. \quad (72)$$

For $m > 0$ and $r \geq 0$, the scalar coefficient is bounded below by m^2 . For $m = 0$ and $r > 0$, it vanishes if and only if $\cos q_1 = \cos q_2 = \cos q_3 = 1$; hence all non-origin cubic corners are lifted. For example, the $(\pi, 0, 0)$ energy squared is exactly $4r^2$. This closes the spectral design problem at the nearest-neighbor Hamiltonian level while preserving the Dirac tangent. It does *not* yet close the stricter QCA problem: $\exp(-itH_W)$ is exactly unitary, but its finite-time exponential is not asserted to be a finite Laurent polynomial or a strictly finite-range one-step update.

Exact temporal blocking and nonclosure. Define the two-step blocking map by $\mathcal{B}(U) = U^2$. The mass-only Plücker subgroup closes exactly,

$$\mathcal{B}(C_z(\varepsilon)) = C_z(2\varepsilon),$$

but the ordered spatial split family does not close by doubling its scalar step. At the exact x - y quarter-turn witness,

$$\mathcal{B}(U_{xy}(\pi/2)) = -\mathbf{1}, \quad U_{xy}(\pi) = \mathbf{1}. \quad (73)$$

The smallest multiplicatively closed enlargement is the submonoid generated by all bare split steps. These statements are NEW **KERNEL**: temporal blocking generates ordered word/coupling data rather than merely a renormalized copy of the original scalar parameters. Spatial decimation is a separate, stronger RG problem.

A proved boundary on scale selection. For the finite positive Plücker action developed here, simultaneous scaling of both primitive spinors by t gives the exact law

$$S(t\psi, t\phi; x) = t^4 S(\psi, \phi; x). \quad (74)$$

Whenever the unscaled action is positive, its radial equation $4t^3 S(\psi, \phi; x) = 0$ is equivalent to $t = 0$. The orthogonal-spinor, positive-quartet control has $S = 1/2$, so the obstruction is nonvacuous. The formal result is now more general. For any positive finite action with radial profile $S(tx) = t^p S(x)$ and natural degree $p > 0$, the derivative at unit scale is $pS(x) > 0$, so the configuration cannot be stationary. At $p = 0$ the radial direction is exactly flat and cannot select an isolated scale. This is NO-GO **KERNEL**: no positive fixed finite homogeneous action can spontaneously select a nonzero scale. A running/refining family, competing homogeneity, constraint, ensemble scale, or other dimensionful input must be displayed. The exact positive control $V_c(t) = (t^2 - c)^2$ has stationary points precisely at $t = 0$ or $t^2 = c$; for supplied $c = 1$, the nonzero points $t = \pm 1$ are global minima. Selection is therefore possible, but its magnitude is exactly the new input. This control is also **REALIZATION KERNEL**.

Exact high-momentum benchmark. The benchmark fixes $z = 3 + 4i$, $\varepsilon = 0.2$, Wilson $r = 1$, and a 10^{-10} kill threshold. The exact-matrix implementation returns three non-origin even-corner aliases (maximum residual 2.30×10^{-16}), both body-center ± 1 modes (residuals below 2.24×10^{-16}), no non-origin Wilson zero, and minimum Wilson corner gap 2. With $r = 0$, all eight corners became gapless as required by the negative control. No parameter was fitted. This numerical result validates the implementation of the theorem chain; the Lean identities, not the benchmark, carry the proof.

Comparison with the tetrahedral walk. Nzongani et al. [?] place four amplitudes on tetrahedral facets, compose facet/strict/weakly local shifts with directional coins, and insert the assigned coin $e^{-i\varepsilon m \gamma^0}$; their expansion recovers the $3 + 1$ Dirac equation to first order. Our register and finite update are different: the present construction is a cubic successive-axis product with a Plücker-derived complex mass coin and an explicit many-step rate. A constant onsite basis change cannot decide away this difference: unitary conjugacy preserves equality of symbols at distinct momenta, so every constant onsite conjugate of our walk inherits the three aliases above and the body-center $\{+1, -1\}$ spectrum. Any equivalence with a tetrahedral update must therefore reproduce those obstructions or use a non-onsite blocking, changed unit cell, or coarse-graining map. A full spectral audit of the tetrahedral symbol is needed before choosing among those possibilities.

Small-step-sensitive $3 + 1$ rate. The live four-factor symbol now has a sharper fixed-time theorem. With $B_4 = |k_x| + |k_y| + |k_z| + |m|$ and $n > 0$,

$$\|U(t/n)^n - e^{-itH}\| \leq \frac{2B_4^2 t^2}{n} \exp\left(\frac{|t|B_4}{n}\right).$$

The proof keeps the small-step exponent through both the split-product remainder and the unitary telescope; unlike the earlier compact-box envelope, it does not replace this factor by $\exp(B_4)$. The resulting growing-window theorem takes $n = K^4$ and permits arbitrary momentum and mass sequences with $B_4 \leq K$; their modewise split-versus-exact error is bounded by $2t^2 K^{-2} \exp(|t|K^{-3})$ and tends to zero. A sharp pure-momentum boundary fixture saturates $B_4 = K$, so this is not a fixed-mode restatement. Sampling and interpolation into physical space, and the position-space L^2 Dirac PDE limit, remain open.

High-symmetry audit of the finite 3+1 split walk

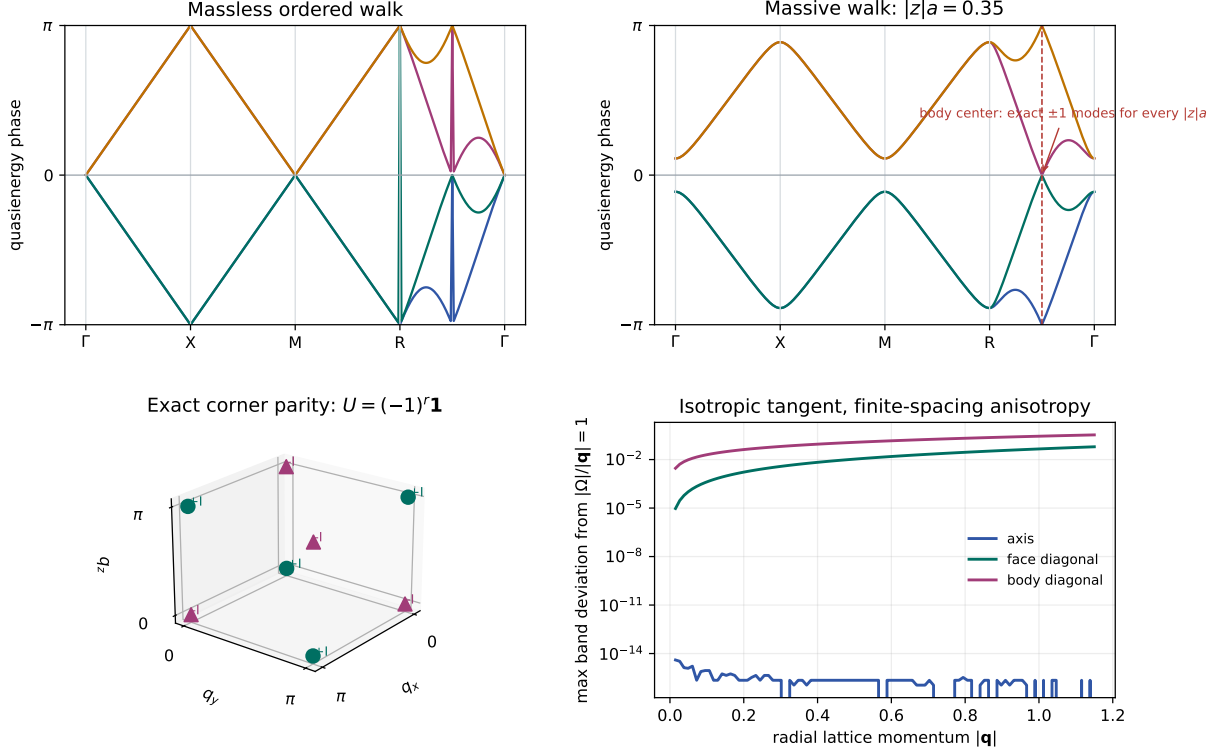


Figure 5: High-symmetry audit of the ordered 3 + 1 walk. The small-momentum tangent is radially isotropic, but the exact massless lattice symbol has four zero-quasienergy corners: the origin and three even-parity aliases. The marked body-center crossing in the massive panel persists for every mass angle by Eq. (??). Curves evaluate the exact finite matrices; the corner classification and body-center eigenmodes are kernel checked.

9 Formal verification and negative controls

The formal development uses Lean 4.28.0 and Mathlib [?, ?]. Its spinor and physics conventions were also cross-checked against the public Physlib corpus (formerly PhysLean and Lean-QuantumInfo) [?]. Formalized mathematical physics at substantial scale is becoming practical: the free bosonic quantum field theory in four-dimensional Euclidean spacetime, satisfying the Glimm–Jaffe axioms, was recently constructed end to end in Lean 4 [?]. The present artifact is complementary in scope: exact finite discrete dynamics, with the physical input — the rest operator itself — derived inside the formalization rather than axiomatized. The trusted Cauchy–Binet identity lives in the project core; the newer operator and walk results remain in the draft namespace while their publication interfaces are reviewed. Each headline declaration below has a build-enforced assumption footprint contained in `{propxt, Classical.choice, Quot.sound}`.

Lean declaration	Mathematical payload
<code>two_edge_plucker_mass_identity</code>	$\det P_{\text{tot}} = \psi \wedge \phi ^2$
<code>pair_diracSymbol_sq</code>	exact Dirac square from the derived mass operator
<code>massOperator_eq_zero_iff_collinear</code>	exact collinearity/gap-closing equivalence
<code>phase_covariance</code>	common mass phase acts by chiral unitary conjugacy
<code>complexPlueckerPathSum_eq_transfer_pow</code>	directed history sum with z and $\bar{z}$
<code>pluecker_positive_mode_chain</code>	the same area fixes the positive rest energy, Euclidean decay, and transfer-pole weight
<code>reflectionPositive_induces_quotient_innerProduct</code>	finite reflection positivity yields a positive physical quotient under explicit compatibility
<code>fixed_time_many_step_bound_on_box</code>	uniform bounded-parameter $O(1/n)$ symbol estimate
<code>phaseDefect_polynomial, gap_zero_iff</code>	transported-phase spectral polynomial and exact zero-mode locus
<code>factorized_degree_one_forced_corner_aliases</code>	no momentum-independent onsite coin removes the tested aliases

The generated artifact supplement retains the complete declaration ledger, negative-control table, and per-module verification commands. They are omitted from the default article view because the scientific evidence is carried by the statements above and the mechanization corrections below.

What mechanization changed

The formal artifact is scientifically useful only where it improves the mathematics. In this development it forced ten corrections that are visible in the paper. First, it separated the first-order rest energy μ from the quadratic invariant $\mu^2 = \det P$ instead of allowing one symbol to drift between operator orders. Second, exact unitarity exposed the finite-spacing corner ratio $\tan(a\mu)$, replacing the tempting but false finite identity $a\mu = \tan(a\mu)$. Third, proving an iterated product estimate separated a one-step Taylor expansion from genuine uniform many-step control. These are not the only bookkeeping gains. Fourth, the exact $3 + 1$ composition preserved the noncommuting $x/y/z/\text{mass}$ factor order instead of identifying reversed local updates. Fifth, the $1 + 1$ full-zone audit separated the zero-quasienergy cone from its high-quasienergy Floquet partner. Sixth, the $3 + 1$ high-symmetry audit disproved the tempting global reading of continuum isotropy: the tangent is isotropic, three nonzero corners exactly alias the origin, and at body center the massive step retains explicit $+1$ and -1 modes for every mass angle. Seventh, the exact Laurent residual ruled out the most economical proposed repair: a nonzero stationary amplitude cannot coexist with origin normalization, exact unitarity, and a full involutory tangent inside one degree-one nearest-neighbor factor. Eighth, the inhomogeneous phase calculation separated a global basis rotation from a genuine endpoint connection and proved that a global real lift cannot carry nonzero winding. Ninth, exact temporal blocking showed that the mass-only group law closes while the noncommuting spatial split generates a larger ordered-word family. Tenth, the radial derivative of the positive Plücker action proved that its homogeneity collapses rather than selects a nonzero scale. These are the formalization contributions; the number of declarations is not.

10 What is forced, derived, realized, and open

The central scientific distinction is not between bold and cautious language. It is between different logical verbs.

Verb	Claim in this paper
Forced	A quadratic invariant vanishing on the full null boundary is proportional to the determinant; a nonnegative first-order gap is proportional to its square root.
Derived	The spinor pair gives z ; z gives B_z ; B_z gives the Dirac square, rest gap, unitary turn gate, and orientation-sensitive corner phase; a site field $z(x)$ gives its own patched link data with an exact 2π -integer winding.
Realized	The split-step null walk exhibits the construction exactly and converges uniformly on bounded $1+1$ momentum boxes; a finite local $3+1$ successive-axis walk has an ordered complex-Plücker symbol with the same explicit rate, a proved quartically growing-window consistency theorem, exact three-torus Plancherel control, and a classified high-symmetry failure set.
Open	Exact identification of the Fourier-conjugate maximal multiplier with the position-space differential Dirac operator and its Sobolev domain, a replacement $3+1$ regulator that removes the corner aliases and body-center modes beyond the excluded degree-one stationary-amplitude ansatz, reduction of the derived link data to a corrected non-winding-only defect index and stable localized mode, instantiation of the proved general free-plus-pair CAR cone with the full live $3+1$ walk, an operator-exponential/action law, and a continuum interaction theory.

What can an assigned scalar mass mimic? Everything that depends on $|z|$ alone: after setting $m = |z|$, the constant-coefficient spectrum, the dispersion, the velocity bound, and the uniform many-step limit are identical, and the paper says so without embarrassment. What it cannot mimic is now a theorem list rather than a narrative. It has no input from which the rest operator is derived rather than declared (the classification theorem makes “derived” precise); it carries one real coin where the exact history sum carries z and $\bar{z}$ on opposite orientations; it produces no endpoint connection, no integer link winding, and no patched transition data from a varying phase unless that field is separately supplied; and a scalar mass, being one real number, cannot reproduce the exact $4/5$ -versus-1 interference split in the supplied pair-kick sector. More importantly, it cannot reproduce the free two-site spectral split between equal-modulus fields without importing exactly the complex datum and its parallel transport that this paper derives. A spatially common constant phase remains removable, as it must; a transported relative phase is already a free-spectrum observable. The pair interaction is a second phase reader, not the condition that makes the free reader exist. Once the phase varies around a cycle, the distinction further becomes patched link data with an integer winding.

There is now also a free two-site spectral consequence. For two equal-modulus rest blocks coupled by the displayed unit edge phase, the exact 4×4 Hamiltonian obeys a scalar quadratic polynomial whose splitting is controlled by the parallel-transported mismatch $\Delta = z_R - (\bar{w})^2 z_L$. Its lower squared-energy level vanishes exactly at $t = |z_L|$ and $z_R = -(\bar{w})^2 z_L$ (for $t \geq 0$), while a common phase on both sites is removed by an explicit unitary conjugacy. The equal-modulus hypothesis is load-bearing: the formal development contains a concrete unequal-modulus counterexample. This is a phase-sensitive finite free spectrum, not yet a topological-protection or localization theorem.

The paper proves a finite free construction and one finite phase-sensitive quartic extension. It does not prove that every massive particle is composite in the ordinary particle-physics sense. The “constituents” here are null momentum factors of a finite spinor description, not additional asymptotic particles. Nor does the theorem compute an absolute mass scale: rescaling the input spinors rescales z and hence μ . The quartic-action no-go sharpens this from an omission to a proved boundary for the natural positive homogeneous action.

There is nevertheless a rigorous finite bridge from a chosen positive Hamiltonian to the usual spectral signatures of mass. For a finite Hermitian $H \geq 0$ and Euclidean spacing $a \geq 0$, the genuine matrix exponential $T = e^{-aH}$ is Hermitian, strictly positive, and contractive. If a nonzero observable vector is an eigenmode with energy $E > 0$, then its transfer orbit is exactly $e^{-aEn}v$, the energy is reconstructed as $-a^{-1} \log(e^{-aE})$, and its scalar transfer resolvent has a unique simple denominator zero with positive coefficient $\|v\|^2$. Every finite orbit kernel is Gram positive. These statements are NEW [KERNEL](#). They show that decay, reconstructed energy, and positive visible pole weight are not independent postulates once the finite Hamiltonian and observable mode are fixed. The Plücker construction now closes the corresponding selected rest-mode chain without an independently assigned mass parameter. For $z = \psi \wedge \phi \neq 0$, the same z that fixes $\det P_{\text{tot}} = |z|^2$ gives an explicit positive-energy eigenvector of B_z with eigenvalue $|z|$. Its Euclidean transfer factor is exactly $e^{-a|z|}$, lies strictly between zero and one for $a > 0$, reconstructs the energy $|z|$, and carries positive visible weight in the scalar transfer denominator. These statements are NEW [KERNEL](#).

The selection is essential rather than cosmetic. The other rest eigenvector has energy $-|z|$ and grows by $e^{a|z|}$ under the same Euclidean step. Thus the full B_z is Hermitian but indefinite; the theorem derives the decaying positive-energy branch, not positivity of the unprojected rest operator. It does not derive an interacting action, an infinite-volume Hamiltonian, or an LSZ particle pole. The “positive coefficient” is the weight recovered after multiplying by the transfer denominator; the complex-analytic residue in a local pole coordinate differs by sign and scale.

The formal development also separates ordinary matrix positivity from finite Osterwalder–Schrader reflection positivity. Under an explicit compatibility condition, reflection positivity makes the reflected positive-time block positive semidefinite and its null-space quotient a genuine real inner-product space. Neither notion implies the other: exact symmetric 2×2 witnesses show both failures of implication. This is NEW [KERNEL](#), but it is the algebraic finite OS step only. Reflection positivity of an interacting lattice gauge-field algebra and reconstruction of its transfer Hamiltonian remain open.

The result does support a precise statement: a timelike rest gap can be encoded entirely by massless rank-one momentum primitives, and the gap is exactly the area opened between their spinor directions. In the derived walk, microscopic translation occurs along null channels while subluminal drift and a rest gap arise from coherent direction changes. This is an exact property of the finite model, not yet a universal ontology of nature.

11 Interpretation and open problems

The larger null-information program studies a four-term finite carrier square, positive cohomology, closure interactions, and discrete frame data. Those structures motivated the present theorem but are not required to prove it. They belong in a companion paper unless they acquire a theorem that depends on B_z .

Three claim-discipline boundaries govern everything in this paper and its companions. First, every invariant here is a finite-dimensional algebraic discriminator on an explicit register; we claim no Fredholm or finite-volume topological index, and any comparison with established real-space walk indices is made through their closed-form evaluation on infinite periodic extensions, never as a finite-volume theorem. Second, the pair interaction is supplied as an even Hermitian four-mode kick consistent with the CAR support and layer schedule; we prove its exact gate, unitarity, and composition behavior, but we do not derive the interaction from a microscopic principle, and the same gap is open in the interacting-automaton literature we compare against. Third, every result in this paper is machine-checked in Lean 4 with its axiom footprint pinned; where companion analyses

use exact computer-algebra computations that are not yet formalized, they are tagged as such there and are never cited here as theorems.

Three frontiers determine whether the construction grows into a physical theory.

1. **Changing-lattice position-space convergence.** Compose the proved symbol estimates, representative-safe cell averaging, ultraviolet tail control, and Fourier identifications into a strong L^2 convergence theorem for the Dirac PDE propagator.
2. **An alias-free exactly local $3 + 1$ unitary regulator.** Escape the proved successive-axis and degree-one onsite obstructions while retaining exact unitarity, finite propagation, the Dirac tangent, and the Plücker-derived complex mass operator.
3. **A derived interacting many-fermion dynamics.** Integrate the exact CAR lift and local pair gates into one field-valued action or update, then obtain a scattering phase, bound state, or selection rule not freely fitted as an input.

The optional research supplement records the finer theorem and construction tasks behind these three gates.

The finite construction encodes spacetime information algebraically through a fixed Pauli map and null direction grading. Emergence of a dynamical continuum geometry remains conjectural. We therefore do not claim that the paper starts without spacetime structure; it starts without primitive timelike motion or a primitive mass term inside the finite model.

12 Conclusion

The central operator theorem is simple. The total momentum P_{tot} determines the radius $|z| = \sqrt{\det P_{\text{tot}}}$ of an odd-Hermitian mass orbit, while an oriented null factorization selects the representative $B_z = (\text{Re } z)\sigma_x - (\text{Im } z)\sigma_y$. This assignment is uniquely fixed within the stated homogeneous and chiral-covariant class. Its positive rest energy is $|z|$, its rest-frame band gap is $2|z|$, and both close exactly when the primitive spinors become collinear.

The orientation survives dynamics in a precise way. B_z exponentiates to an exact unitary turn, opposite histories carry z and $\bar{z}$, and the walk has a uniform $O(1/n)$ continuum-symbol estimate. A common phase is removable, but transport between local chiral frames leaves the invariant mismatch $|z_R - (\bar{w})^2 z_L|$, which splits the exact two-site free spectrum. An assigned complex mass can reproduce that phase physics; the contribution of the Plücker construction is to derive the complex field, its transport law, its determinant relation, and its zero locus from one null-spinor input.

The same exact treatment identifies the regulator boundary. The tested successive-axis $3 + 1$ extension retains classified zero- and π -quasienergy aliases, and no momentum-independent onsite coin removes them. The decisive frontiers are therefore an explicit changing-lattice position-space Dirac limit, an alias-free exactly local $3 + 1$ unitary regulator, and a derived interacting many-fermion dynamics. These are separate mathematical tests of whether the finite mass-orbit construction can become a physical theory.

References

- [1] H. Elvang and Y.-t. Huang, *Scattering Amplitudes*, arXiv:1308.1697.

- [2] N. Arkani-Hamed, T.-C. Huang, and Y.-t. Huang, *Scattering Amplitudes For All Masses and Spins*, JHEP 11 (2021) 070, arXiv:1709.04891.
- [3] S. Dittmaier, *Weyl–van der Waerden formalism for helicity amplitudes of massive particles*, Phys. Rev. D 59 (1998) 016007, arXiv:hep-ph/9805445.
- [4] R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals*, McGraw–Hill, 1965.
- [5] T. Jacobson and L. S. Schulman, *Quantum stochasticity: the passage from a relativistic to a non-relativistic path integral*, J. Phys. A 17 (1984) 375.
- [6] A. Bisio, G. M. D’Ariano, and A. Tosini, *Quantum field as a quantum cellular automaton: the Dirac free evolution in one dimension*, Ann. Phys. 354 (2015) 244, arXiv:1212.2839.
- [7] G. M. D’Ariano, N. Mosco, P. Perinotti, and A. Tosini, *Path-integral solution of the one-dimensional Dirac quantum cellular automaton*, Phys. Lett. A 378 (2014) 3165–3168, arXiv:1406.1021.
- [8] A. Mallick and C. M. Chandrashekar, *Dirac cellular automaton from split-step quantum walk*, Sci. Rep. 6 (2016) 25779, arXiv:1509.08851.
- [9] S. Succi, F. Fillion-Gourdeau, and S. Palpacelli, *Quantum lattice Boltzmann is a quantum walk*, EPJ Quantum Technol. 2 (2015) 12.
- [10] A. Bisio, G. M. D’Ariano, P. Perinotti, and A. Tosini, *Free quantum field theory from quantum cellular automata: Derivation of Weyl, Dirac and Maxwell quantum cellular automata*, Found. Phys. 46 (2016) 1032–1081, arXiv:1601.04832.
- [11] U. Nsongani, N. Eon, I. Marquez-Martin, A. Perez, G. Di Molfetta, and P. Arrighi, *Dirac quantum walk on tetrahedra*, Phys. Rev. A 110 (2024) 042418, arXiv:2404.09840.
- [12] J. K. Asboth and H. Obuse, *Bulk-boundary correspondence for chiral symmetric quantum walks*, Phys. Rev. B 88 (2013) 121406(R), arXiv:1303.1199.
- [13] D. Gross, V. Nesme, H. Vogts, and R. F. Werner, *Index theory of one dimensional quantum walks and cellular automata*, Commun. Math. Phys. 310 (2012) 419–454, arXiv:0910.3675.
- [14] N. Read, *Compactly-supported Wannier functions and algebraic K-theory*, Phys. Rev. B 95 (2017) 115309, arXiv:1608.04696.
- [15] T. Bessho and M. Sato, *Nielsen–Ninomiya theorem with bulk topology: Duality in Floquet and non-Hermitian systems*, Phys. Rev. Lett. 127 (2021) 196404, arXiv:2006.04204.
- [16] C. Gupta and A. J. Short, *Fermion doubling in Dirac quantum walks*, Phys. Rev. A 114 (2026) 012208, arXiv:2601.15885, doi:10.1103/kkz3-fthm.
- [17] K. Bakircioglu and P. Arnault, *Fermion-doubling problem in chiral discretizations of quantum field theory: Definitive proof, fixing, and computation of two-point correlation function*, arXiv:2607.14874 (2026).
- [18] P. Arnault and F. Debbasch, *Quantum walks and discrete gauge theories*, Phys. Rev. A 93 (2016) 052301, arXiv:1508.00038.

- [19] J. Goldstone and F. Wilczek, *Fractional quantum numbers on solitons*, Phys. Rev. Lett. 47 (1981) 986–989, doi:10.1103/PhysRevLett.47.986.
- [20] H. B. Nielsen and M. Ninomiya, *Absence of neutrinos on a lattice: (I). Proof by homotopy theory*, Nucl. Phys. B (1981), doi:10.1016/0550-3213(82)90011-6; *(II). Intuitive topological proof*, Nucl. Phys. B (1981), doi:10.1016/0550-3213(81)90524-1.
- [21] L. Mlodinow and T. A. Brun, *Discrete spacetime, quantum walks, and relativistic wave equations*, Phys. Rev. A 97 (2018) 042131, arXiv:1802.03910.
- [22] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, *A theory of Trotter error*, Phys. Rev. X 11 (2021) 011020, arXiv:1912.08854.
- [23] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, Lecture Notes in Computer Science 12699 (2021) 625–635.
- [24] The Mathlib Community, *The Lean mathematical library*, Proceedings of CPP 2020, 367–381.
- [25] The Physlib Community, *Physlib: Digitalizing physics in Lean 4*, formerly PhysLean and Lean-QuantumInfo, <https://physlib.io/>; see also J. Tooby-Smith, *HepLean: Digitalising high energy physics*, arXiv:2405.08863.
- [26] M. R. Douglas, S. Hoback, A. Mei, and R. Nissim, *Formalization of QFT*, arXiv:2603.15770.